US012414811B2

(12) **United States Patent**
Childs et al.

(10) **Patent No.: US 12,414,811 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **DEVICES AND METHODS FOR LEAFLET CUTTING**

(71) Applicant: **Evalve, Inc.**, Santa Clara, CA (US)

(72) Inventors: **Richard Thomas Childs**, San Francisco, CA (US); **Scott C. Mosher**, San Francisco, CA (US); **Lauren Troxler Harvey**, Austin, TX (US); **Koji J. Kizuka**, Redwood City, CA (US); **Ajitkumar B. Nair**, Pleasanton, CA (US); **Laura M. Kalvass**, Mountain View, CA (US)

(73) Assignee: **Evalve, Inc.**, Santa Clara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 22 days.

(21) Appl. No.: **17/306,398**

(22) Filed: **May 3, 2021**

(65) **Prior Publication Data**

US 2021/0346081 A1 Nov. 11, 2021

Related U.S. Application Data

(60) Provisional application No. 63/020,669, filed on May 6, 2020.

(51) **Int. Cl.**
A61B 18/00 (2006.01)
A61B 18/14 (2006.01)

(52) **U.S. Cl.**
CPC ***A61B 18/1442*** (2013.01); ***A61B 18/1492*** (2013.01); *A61B 2018/00369* (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A61B 18/1442; A61B 18/1492; A61B 2018/00369; A61B 2018/1452; A61B 2017/00783
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,996,261	A	4/1935	Storz
2,097,018	A	10/1937	Chamberlin

(Continued)

FOREIGN PATENT DOCUMENTS

CN	1469724	A	1/2004
CN	102770080	A	11/2012

(Continued)

OTHER PUBLICATIONS

Kruger et al., "P73—Edge to Edge Technique in Complex Mitral Valve Repair," Thorac Cardiovasc Surg., 48(Suppl. 1):106 (2000).

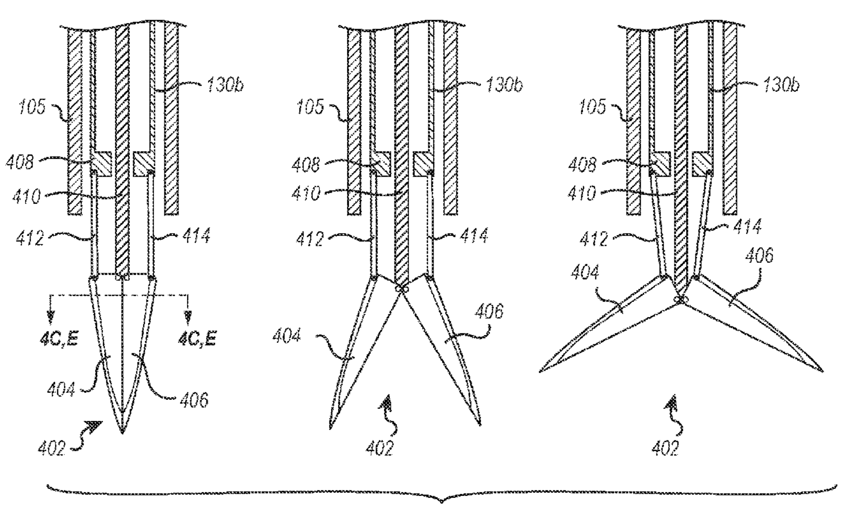
(Continued)

Primary Examiner — Joseph A Stoklosa
Assistant Examiner — Annie L Shoulders
(74) *Attorney, Agent, or Firm* — Workman Nydegger

(57) **ABSTRACT**

A system for cutting leaflet tissue at a cardiac valve may comprise a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve. The system may also include a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve. Finally, the system may comprise a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to cut a portion of leaflet tissue of the cardiac valve.

21 Claims, 16 Drawing Sheets



(52) **U.S. Cl.**
CPC *A61B 2018/0072* (2013.01); *A61B 2018/1452* (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,108,206 A 2/1938 Mecker
3,296,668 A 1/1967 Aiken
3,378,010 A 4/1968 Codling et al.
3,470,875 A 10/1969 Johnson
3,557,780 A 1/1971 Sato
3,671,979 A 6/1972 Moulopoulos
3,675,639 A 7/1972 Cimber
3,776,237 A 12/1973 Hill et al.
3,874,338 A 4/1975 Happel
3,874,388 A 4/1975 King et al.
4,007,743 A 2/1977 Blake
4,056,854 A 11/1977 Boretos et al.
4,064,881 A 12/1977 Meredith
4,091,815 A 5/1978 Larsen
4,112,951 A 9/1978 Hulka et al.
4,235,238 A 11/1980 Ogiu et al.
4,297,749 A 11/1981 Davis et al.
4,312,337 A 1/1982 Donohue
4,425,908 A 1/1984 Simon
4,458,682 A 7/1984 Cerwin
4,484,579 A 11/1984 Meno et al.
4,487,205 A 12/1984 Di et al.
4,498,476 A 2/1985 Cerwin et al.
4,510,934 A 4/1985 Batra
4,531,522 A 7/1985 Bedi et al.
4,578,061 A 3/1986 Lemelson
4,641,366 A 2/1987 Yokoyama et al.
4,686,965 A 8/1987 Bonnet et al.
4,777,951 A 10/1988 Cribier et al.
4,809,695 A 3/1989 Gwathmey et al.
4,872,455 A 10/1989 Pinchuk et al.
4,878,495 A 11/1989 Grayzel
4,917,089 A 4/1990 Sideris
4,944,295 A 7/1990 Gwathmey et al.
4,969,890 A 11/1990 Sugita et al.
4,994,077 A 2/1991 Dobben
5,015,249 A 5/1991 Nakao et al.
5,019,096 A 5/1991 Fox et al.
5,042,707 A 8/1991 Taheri
5,047,041 A 9/1991 Samuels
5,049,153 A 9/1991 Nakao et al.
5,053,043 A 10/1991 Gottesman et al.
5,061,277 A 10/1991 Carpentier et al.
5,069,679 A 12/1991 Taheri
5,071,428 A 12/1991 Chin et al.
5,078,722 A 1/1992 Stevens
5,078,723 A 1/1992 Dance et al.
5,108,368 A 4/1992 Hammerslag et al.
5,125,758 A 6/1992 Dewan
5,171,252 A 12/1992 Friedland
5,171,259 A 12/1992 Inoue
5,190,554 A 3/1993 Coddington et al.
5,195,968 A 3/1993 Lundquist et al.
5,209,756 A 5/1993 Seedhom et al.
5,217,460 A 6/1993 Knoepfler
5,226,429 A 7/1993 Kuzmak
5,226,911 A 7/1993 Chee et al.
5,234,437 A 8/1993 Sepetka
5,242,456 A 9/1993 Nash et al.
5,250,071 A 10/1993 Palermo
5,251,611 A 10/1993 Zehel et al.
5,254,130 A 10/1993 Poncet et al.
5,261,916 A 11/1993 Engelson
5,271,381 A 12/1993 Ailinger et al.
5,275,578 A 1/1994 Adams
5,282,845 A 2/1994 Bush et al.
5,304,131 A 4/1994 Paskar
5,306,283 A 4/1994 Conners
5,306,286 A 4/1994 Stack et al.
5,312,415 A 5/1994 Palermo
5,314,424 A 5/1994 Nicholas
5,318,525 A 6/1994 West et al.
5,320,632 A 6/1994 Heidmueller
5,325,845 A 7/1994 Adair
5,330,442 A 7/1994 Green et al.
5,332,402 A 7/1994 Teitelbaum
5,336,227 A 8/1994 Nakao et al.
5,342,393 A 8/1994 Stack
5,350,397 A 9/1994 Palermo et al.
5,350,399 A 9/1994 Erlebacher et al.
5,359,994 A 11/1994 Krauter et al.
5,368,564 A 11/1994 Savage
5,368,601 A 11/1994 Sauer et al.
5,373,854 A 12/1994 Kolozsi
5,383,886 A 1/1995 Kensey et al.
5,387,219 A 2/1995 Rappe
5,391,182 A 2/1995 Chin
5,395,030 A 3/1995 Kuramoto et al.
5,403,312 A 4/1995 Yates et al.
5,403,326 A 4/1995 Harrison et al.
5,411,552 A 5/1995 Andersen et al.
5,417,684 A 5/1995 Jackson et al.
5,417,699 A 5/1995 Klein et al.
5,417,700 A 5/1995 Egan
5,423,830 A 6/1995 Schneebaum et al.
5,423,857 A 6/1995 Rosenman et al.
5,423,858 A 6/1995 Bolanos et al.
5,423,882 A 6/1995 Jackman et al.
5,431,666 A 7/1995 Sauer et al.
5,437,551 A 8/1995 Chalifoux
5,437,681 A 8/1995 Meade et al.
5,447,966 A 9/1995 Hermes et al.
5,450,860 A 9/1995 O'Connor
5,456,400 A 10/1995 Shichman et al.
5,456,684 A 10/1995 Schmidt et al.
5,462,527 A 10/1995 Stevens-Wright et al.
5,472,044 A 12/1995 Hall et al.
5,472,423 A 12/1995 Gronauer
5,476,470 A 12/1995 Fitzgibbons, Jr.
5,477,856 A 12/1995 Lundquist
5,478,309 A 12/1995 Sweezer et al.
5,478,353 A 12/1995 Yoon
5,487,746 A 1/1996 Yu et al.
5,496,332 A 3/1996 Sierra et al.
5,507,725 A 4/1996 Savage et al.
5,507,755 A 4/1996 Gresl et al.
5,507,757 A 4/1996 Sauer et al.
5,520,701 A 5/1996 Lerch
5,522,873 A 6/1996 Jackman et al.
5,527,313 A 6/1996 Scott et al.
5,527,321 A 6/1996 Hinchliffe
5,527,322 A 6/1996 Klein et al.
5,536,251 A 7/1996 Evard et al.
5,540,705 A 7/1996 Meade et al.
5,542,949 A 8/1996 Yoon
5,554,185 A 9/1996 Block et al.
5,562,678 A 10/1996 Booker
5,569,274 A 10/1996 Rapacki et al.
5,571,085 A 11/1996 Accisano, III
5,571,137 A 11/1996 Marlow et al.
5,571,215 A 11/1996 Sterman et al.
5,575,802 A 11/1996 McQuilkin et al.
5,582,611 A 12/1996 Tsuruta et al.
5,584,803 A * 12/1996 Stevens A61M 25/0147 604/6.16
5,593,424 A 1/1997 Northrup, III
5,593,435 A 1/1997 Carpentier et al.
5,609,598 A 3/1997 Laufer et al.
5,617,854 A 4/1997 Munsif
5,618,306 A 4/1997 Roth et al.
5,620,452 A 4/1997 Yoon
5,620,461 A 4/1997 Muijs et al.
5,626,588 A 5/1997 Sauer et al.
5,634,932 A 6/1997 Schmidt
5,636,634 A 6/1997 Kordis et al.
5,639,277 A 6/1997 Mariant et al.
5,640,955 A 6/1997 Ockuly et al.
5,649,937 A 7/1997 Bito et al.
5,662,681 A 9/1997 Nash et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,669,917	A	9/1997	Sauer et al.
5,669,919	A	9/1997	Sanders et al.
5,690,671	A	11/1997	McGurk et al.
5,695,504	A	12/1997	Gifford et al.
5,695,505	A	12/1997	Yoon
5,702,825	A	12/1997	Keita et al.
5,706,824	A	1/1998	Whittier
5,709,707	A	1/1998	Lock et al.
5,713,910	A	2/1998	Gordon et al.
5,713,911	A	2/1998	Racenet et al.
5,715,817	A	2/1998	Stevens-Wright et al.
5,716,367	A	2/1998	Koike et al.
5,718,725	A	2/1998	Sterman et al.
5,719,725	A	2/1998	Nakao
5,722,421	A	3/1998	Francese et al.
5,725,542	A	3/1998	Yoon
5,725,556	A	3/1998	Moser et al.
5,738,649	A	4/1998	Macoviak
5,741,271	A	4/1998	Nakao et al.
5,741,280	A	4/1998	Fleenor
5,746,747	A	5/1998	McKeating
5,749,828	A	5/1998	Yeung
5,759,193	A	6/1998	Burbank et al.
5,769,812	A	6/1998	Stevens et al.
5,769,863	A	6/1998	Garrison
5,772,578	A	6/1998	Heimberger et al.
5,782,845	A	7/1998	Shewchuk
5,797,927	A	8/1998	Yoon
5,797,960	A	8/1998	Stevens et al.
5,810,847	A	9/1998	Laufer et al.
5,810,849	A	9/1998	Kontos
5,810,853	A	9/1998	Yoon
5,810,876	A	9/1998	Kelleher
5,814,029	A	9/1998	Hassett
5,820,591	A	10/1998	Thompson et al.
5,820,592	A	10/1998	Hammerslag
5,820,630	A	10/1998	Lind
5,820,631	A	10/1998	Nobles
5,823,955	A	10/1998	Kuck et al.
5,823,956	A	10/1998	Roth et al.
5,824,065	A	10/1998	Gross
5,827,237	A	10/1998	Macoviak et al.
5,829,447	A	11/1998	Stevens et al.
5,833,671	A	11/1998	Macoviak et al.
5,836,955	A	11/1998	Buelna et al.
5,840,081	A	11/1998	Andersen et al.
5,843,031	A	12/1998	Hermann et al.
5,843,103	A	12/1998	Wulfman
5,849,019	A	12/1998	Yoon
5,853,422	A	12/1998	Huebsch et al.
5,855,271	A	1/1999	Eubanks et al.
5,855,590	A	1/1999	Malecki et al.
5,855,614	A	1/1999	Stevens et al.
5,860,990	A	1/1999	Nobles et al.
5,861,003	A	1/1999	Latson et al.
5,868,733	A	2/1999	Ockuly et al.
5,876,399	A	3/1999	Chia et al.
5,879,307	A	3/1999	Chio et al.
5,885,271	A	3/1999	Hamilton et al.
5,891,160	A	4/1999	Williamson et al.
5,895,404	A	4/1999	Ruiz
5,895,417	A	4/1999	Pomeranz et al.
5,906,620	A	5/1999	Nakao et al.
5,908,420	A	6/1999	Parins et al.
5,916,147	A	6/1999	Boury
5,928,224	A	7/1999	Laufer
5,944,733	A	8/1999	Engelson
5,947,363	A	9/1999	Bolduc et al.
5,954,732	A	9/1999	Hart et al.
5,957,949	A	9/1999	Leonhardt et al.
5,957,973	A	9/1999	Quiachon et al.
5,972,020	A	10/1999	Carpentier et al.
5,972,030	A	10/1999	Garrison et al.
5,980,455	A	11/1999	Daniel et al.
5,989,284	A	11/1999	Laufer
5,997,547	A	12/1999	Nakao et al.
6,007,546	A	12/1999	Snow et al.
6,015,417	A	1/2000	Reynolds, Jr.
6,019,722	A	2/2000	Spence et al.
6,022,360	A	2/2000	Reimels et al.
6,033,378	A	3/2000	Lundquist et al.
6,033,419	A	3/2000	Hamblin et al.
6,036,699	A	3/2000	Andreas et al.
6,048,351	A	4/2000	Gordon et al.
6,053,933	A	4/2000	Balazs et al.
6,056,769	A	5/2000	Epstein et al.
6,059,757	A	5/2000	Macoviak et al.
6,060,628	A	5/2000	Aoyama et al.
6,060,629	A	5/2000	Pham et al.
6,063,106	A	5/2000	Gibson
6,066,146	A	5/2000	Carroll et al.
6,068,628	A	5/2000	Fanton et al.
6,068,629	A	5/2000	Haissaguerre et al.
6,077,214	A	6/2000	Mortier et al.
6,086,600	A	7/2000	Kortenbach
6,088,889	A	7/2000	Luther et al.
6,090,118	A	7/2000	McGuckin, Jr.
6,099,505	A	8/2000	Ryan et al.
6,099,553	A	8/2000	Hart et al.
6,110,145	A	8/2000	Macoviak
6,117,144	A	9/2000	Nobles et al.
6,117,159	A	9/2000	Huebsch et al.
6,123,665	A	9/2000	Kawano
6,123,699	A	9/2000	Webster, Jr.
6,126,658	A	10/2000	Baker
6,132,447	A	10/2000	Dorsey
6,136,010	A	10/2000	Modesitt et al.
6,139,508	A	10/2000	Simpson et al.
6,143,024	A	11/2000	Campbell et al.
6,159,240	A	12/2000	Sparer et al.
6,162,233	A	12/2000	Williamson et al.
6,165,164	A	12/2000	Hill et al.
6,165,183	A	12/2000	Kuehn et al.
6,165,204	A	12/2000	Levinson et al.
6,168,614	B1	1/2001	Andersen et al.
6,171,320	B1	1/2001	Monassevitch
6,174,322	B1	1/2001	Schneidt
6,180,059	B1	1/2001	Divino et al.
6,182,664	B1	2/2001	Cosgrove
6,187,003	B1	2/2001	Buysse et al.
6,190,408	B1	2/2001	Melvin
6,197,043	B1	3/2001	Davidson
6,203,531	B1	3/2001	Ockuly et al.
6,203,553	B1	3/2001	Robertson et al.
6,206,893	B1	3/2001	Klein et al.
6,206,907	B1	3/2001	Marino et al.
6,210,419	B1	4/2001	Mayenberger et al.
6,210,432	B1	4/2001	Solem et al.
6,245,079	B1	6/2001	Nobles et al.
6,264,617	B1	7/2001	Bales et al.
6,267,746	B1	7/2001	Bumbalough
6,267,781	B1	7/2001	Tu
6,269,819	B1	8/2001	Oz et al.
6,277,555	B1	8/2001	Duran et al.
6,283,127	B1	9/2001	Sterman et al.
6,283,962	B1	9/2001	Tu et al.
6,299,637	B1	10/2001	Shaolian et al.
6,306,133	B1	10/2001	Tu et al.
6,312,447	B1	11/2001	Grimes
6,319,250	B1	11/2001	Falwell et al.
6,322,559	B1	11/2001	Daulton et al.
6,332,893	B1	12/2001	Mortier et al.
6,334,860	B1	1/2002	Dorn
6,352,708	B1	3/2002	Duran et al.
6,355,030	B1	3/2002	Aldrich et al.
6,358,277	B1	3/2002	Duran
6,368,326	B1	4/2002	Dakin et al.
6,387,104	B1	5/2002	Pugsley et al.
6,402,780	B2	6/2002	Williamson et al.
6,402,781	B1	6/2002	Langberg et al.
6,406,420	B1	6/2002	McCarthy et al.
6,419,640	B1	7/2002	Taylor
6,419,669	B1	7/2002	Frazier et al.
6,461,366	B1	10/2002	Seguin

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,464,707	B1	10/2002	Bjerken
6,482,224	B1	11/2002	Michler et al.
6,485,489	B2	11/2002	Teirstein et al.
6,494,881	B1	12/2002	Bales et al.
6,508,828	B1	1/2003	Akerfeldt et al.
6,517,550	B1	2/2003	Konya et al.
6,533,796	B1	3/2003	Sauer et al.
6,537,314	B2	3/2003	Langberg et al.
6,540,755	B2	4/2003	Ockuly et al.
6,551,331	B2	4/2003	Nobles et al.
6,562,037	B2	5/2003	Paton et al.
6,562,052	B2	5/2003	Nobles et al.
6,575,971	B2	6/2003	Hauck et al.
6,585,761	B2	7/2003	Taheri
6,599,311	B1	7/2003	Biggs et al.
6,616,684	B1	9/2003	Vidlund et al.
6,619,291	B2	9/2003	Hlavka et al.
6,626,899	B2	9/2003	Houser et al.
6,626,921	B2	9/2003	Blatter et al.
6,626,930	B1	9/2003	Allen et al.
6,629,534	B1	10/2003	St et al.
6,641,592	B1	11/2003	Sauer et al.
6,656,221	B2	12/2003	Taylor et al.
6,669,687	B1	12/2003	Saadat
6,685,648	B2	2/2004	Flaherty et al.
6,689,164	B1	2/2004	Seguin
6,695,866	B1	2/2004	Kuehn et al.
6,701,929	B2	3/2004	Hussein
6,702,825	B2	3/2004	Frazier et al.
6,702,826	B2	3/2004	Liddicoat et al.
6,709,382	B1	3/2004	Horner
6,709,456	B2	3/2004	Langberg et al.
6,718,985	B2	4/2004	Hlavka et al.
6,719,767	B1	4/2004	Kimblad
6,723,038	B1	4/2004	Schroeder et al.
6,726,716	B2	4/2004	Marquez
6,740,107	B2	5/2004	Loeb et al.
6,746,471	B2	6/2004	Mortier et al.
6,752,813	B2	6/2004	Goldfarb et al.
6,755,777	B2	6/2004	Schweich et al.
6,764,510	B2	7/2004	Vidlund et al.
6,767,349	B2	7/2004	Ouchi
6,770,083	B2	8/2004	Seguin
6,797,001	B2	9/2004	Mathis et al.
6,797,002	B2	9/2004	Spence et al.
6,860,179	B2	3/2005	Hopper et al.
6,875,224	B2	4/2005	Grimes
6,926,715	B1	8/2005	Hauck et al.
6,932,810	B2	8/2005	Ryan
6,945,978	B1	9/2005	Hyde
6,949,122	B2	9/2005	Adams et al.
6,966,914	B2	11/2005	Abe
6,986,775	B2	1/2006	Morales et al.
7,004,970	B2	2/2006	Cauthen et al.
7,011,669	B2	3/2006	Kimblad
7,033,390	B2	4/2006	Johnson et al.
7,048,754	B2	5/2006	Martin et al.
7,056,294	B2 *	6/2006	Khairkhahan ... A61B 17/32075 606/167
7,112,207	B2	9/2006	Allen et al.
7,226,467	B2	6/2007	Lucatero et al.
7,258,694	B1	8/2007	Choi et al.
7,288,097	B2	10/2007	Seguin
7,291,168	B2	11/2007	Macoviak et al.
7,338,467	B2	3/2008	Lutter
7,381,210	B2	6/2008	Zarbatany et al.
7,435,257	B2	10/2008	Lashinski et al.
7,464,712	B2	12/2008	Oz et al.
7,497,822	B1	3/2009	Kugler et al.
7,533,790	B1	5/2009	Knodel et al.
7,563,267	B2	7/2009	Goldfarb et al.
7,563,273	B2	7/2009	Goldfarb et al.
7,604,646	B2	10/2009	Goldfarb et al.
7,608,091	B2	10/2009	Goldfarb et al.
7,635,329	B2	12/2009	Goldfarb et al.
7,651,502	B2	1/2010	Jackson
7,655,015	B2	2/2010	Goldfarb et al.
7,666,204	B2	2/2010	Thornton et al.
7,955,340	B2	6/2011	Michlitsch et al.
8,216,234	B2	7/2012	Long
8,257,356	B2	9/2012	Bleich et al.
8,398,708	B2	3/2013	Meiri et al.
8,435,237	B2	5/2013	Bahney
8,496,655	B2	7/2013	Epp et al.
8,500,768	B2	8/2013	Cohen
8,523,881	B2	9/2013	Cabiri et al.
8,623,077	B2	1/2014	Cohn
8,690,858	B2	4/2014	Machold et al.
8,821,518	B2	9/2014	Saliman et al.
8,926,588	B2	1/2015	Berthiaume et al.
9,126,032	B2	9/2015	Khairkhahan et al.
9,211,119	B2	12/2015	Hendricksen et al.
9,370,341	B2	6/2016	Ceniccola et al.
9,498,331	B2	11/2016	Chang et al.
9,572,666	B2	2/2017	Basude et al.
9,770,256	B2	9/2017	Cohen et al.
9,949,833	B2	4/2018	McCleary et al.
10,238,493	B1 *	3/2019	Metchik A61F 2/2466
10,667,804	B2	6/2020	Basude et al.
11,013,554	B2	5/2021	Coates
11,406,250	B2	8/2022	Saadat et al.
2001/0002445	A1	5/2001	Vesely
2001/0004715	A1	6/2001	Duran et al.
2001/0005787	A1	6/2001	Oz et al.
2001/0010005	A1	7/2001	Kammerer et al.
2001/0018611	A1	8/2001	Solem et al.
2001/0022872	A1	9/2001	Marui
2001/0037084	A1	11/2001	Nardeo
2001/0039411	A1	11/2001	Johansson et al.
2001/0044568	A1	11/2001	Langberg et al.
2001/0044635	A1	11/2001	Niizeki et al.
2002/0013547	A1	1/2002	Paskar
2002/0013571	A1	1/2002	Goldfarb et al.
2002/0022848	A1	2/2002	Garrison et al.
2002/0026233	A1	2/2002	Shaknovich
2002/0035361	A1	3/2002	Houser et al.
2002/0035381	A1	3/2002	Bardy et al.
2002/0042651	A1	4/2002	Liddicoat et al.
2002/0055767	A1	5/2002	Forde et al.
2002/0055774	A1	5/2002	Liddicoat
2002/0055775	A1	5/2002	Carpentier et al.
2002/0058910	A1	5/2002	Hermann et al.
2002/0058995	A1	5/2002	Stevens
2002/0077687	A1	6/2002	Ahn
2002/0087148	A1	7/2002	Brock et al.
2002/0087169	A1	7/2002	Brock et al.
2002/0087173	A1	7/2002	Alferness et al.
2002/0103532	A1	8/2002	Langberg et al.
2002/0107534	A1	8/2002	Schaefer et al.
2002/0147456	A1	10/2002	Diduch et al.
2002/0156526	A1	10/2002	Hlavka et al.
2002/0158528	A1	10/2002	Tsuzaki et al.
2002/0161378	A1	10/2002	Downing
2002/0169360	A1	11/2002	Taylor et al.
2002/0173811	A1	11/2002	Tu et al.
2002/0173841	A1	11/2002	Ortiz et al.
2002/0183766	A1	12/2002	Seguin
2002/0183787	A1	12/2002	Wahr et al.
2002/0183835	A1	12/2002	Taylor et al.
2003/0005797	A1	1/2003	Hopper et al.
2003/0045778	A1	3/2003	Ohline et al.
2003/0050693	A1	3/2003	Quijano et al.
2003/0069570	A1	4/2003	Witzel et al.
2003/0069593	A1	4/2003	Tremulis et al.
2003/0069636	A1	4/2003	Solem et al.
2003/0074012	A1	4/2003	Nguyen et al.
2003/0078654	A1	4/2003	Taylor et al.
2003/0083742	A1	5/2003	Spence et al.
2003/0105519	A1	6/2003	Fasol et al.
2003/0105520	A1	6/2003	Alferness et al.
2003/0120340	A1	6/2003	Liska et al.
2003/0120341	A1	6/2003	Shennib et al.
2003/0130669	A1	7/2003	Damarati
2003/0130730	A1	7/2003	Cohn et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2003/0144697	A1	7/2003	Mathis et al.	
2003/0167071	A1	9/2003	Martin et al.	
2003/0171776	A1	9/2003	Adams et al.	
2003/0187467	A1	10/2003	Schreck	
2003/0195562	A1	10/2003	Collier et al.	
2003/0208231	A1	11/2003	Williamson et al.	
2003/0229395	A1	12/2003	Cox	
2003/0233038	A1	12/2003	Hassett	
2004/0002719	A1	1/2004	Oz et al.	
2004/0003819	A1	1/2004	St et al.	
2004/0015232	A1	1/2004	Shu et al.	
2004/0019377	A1	1/2004	Taylor et al.	
2004/0019378	A1	1/2004	Hlavka et al.	
2004/0024414	A1	2/2004	Downing	
2004/0030319	A1*	2/2004	Korkor	A61M 25/0074 604/506
2004/0030382	A1	2/2004	St et al.	
2004/0034380	A1	2/2004	Woolfson et al.	
2004/0039442	A1	2/2004	St et al.	
2004/0039443	A1	2/2004	Solem et al.	
2004/0044350	A1	3/2004	Martin et al.	
2004/0044365	A1	3/2004	Bachman	
2004/0049207	A1	3/2004	Goldfarb et al.	
2004/0049211	A1	3/2004	Tremulis et al.	
2004/0059345	A1	3/2004	Nakao et al.	
2004/0073302	A1	4/2004	Rourke et al.	
2004/0078053	A1	4/2004	Berg et al.	
2004/0087975	A1	5/2004	Lucatero et al.	
2004/0088047	A1	5/2004	Spence et al.	
2004/0092858	A1	5/2004	Wilson et al.	
2004/0092962	A1	5/2004	Thornton et al.	
2004/0097878	A1	5/2004	Anderson et al.	
2004/0097979	A1	5/2004	Svanidze et al.	
2004/0106989	A1	6/2004	Wilson et al.	
2004/0111099	A1	6/2004	Nguyen et al.	
2004/0116848	A1	6/2004	Gardeski et al.	
2004/0116951	A1	6/2004	Rosengart	
2004/0122448	A1	6/2004	Levine	
2004/0127849	A1	7/2004	Kantor	
2004/0127981	A1	7/2004	Rahdert et al.	
2004/0127982	A1	7/2004	Machold et al.	
2004/0127983	A1	7/2004	Mortier et al.	
2004/0133062	A1	7/2004	Pai et al.	
2004/0133063	A1	7/2004	McCarthy et al.	
2004/0133082	A1	7/2004	Abraham-Fuchs et al.	
2004/0133192	A1	7/2004	Houser et al.	
2004/0133220	A1	7/2004	Lashinski et al.	
2004/0133232	A1	7/2004	Rosenbluth et al.	
2004/0133240	A1	7/2004	Adams et al.	
2004/0133273	A1	7/2004	Cox	
2004/0138744	A1	7/2004	Lashinski et al.	
2004/0138745	A1	7/2004	Macoviak et al.	
2004/0147826	A1	7/2004	Peterson	
2004/0148021	A1	7/2004	Cartledge et al.	
2004/0152847	A1	8/2004	Emri et al.	
2004/0152947	A1	8/2004	Schroeder et al.	
2004/0153144	A1	8/2004	Seguin	
2004/0158123	A1	8/2004	Jayaraman	
2004/0162610	A1	8/2004	Liska et al.	
2004/0167539	A1	8/2004	Kuehn et al.	
2004/0186486	A1	9/2004	Roue et al.	
2004/0186566	A1	9/2004	Hindrichs et al.	
2004/0193191	A1	9/2004	Starksen et al.	
2004/0215339	A1	10/2004	Drasler et al.	
2004/0220593	A1	11/2004	Greenhalgh	
2004/0220657	A1	11/2004	Nieminen et al.	
2004/0225233	A1	11/2004	Frankowski et al.	
2004/0225300	A1	11/2004	Goldfarb et al.	
2004/0225305	A1	11/2004	Ewers et al.	
2004/0225353	A1	11/2004	McGuckin et al.	
2004/0236354	A1	11/2004	Seguin	
2004/0242960	A1	12/2004	Orban	
2004/0243229	A1	12/2004	Vidlund et al.	
2004/0249452	A1	12/2004	Adams et al.	
2004/0249453	A1	12/2004	Cartledge et al.	
2004/0260393	A1	12/2004	Rahdert et al.	
2005/0004583	A1	1/2005	Oz et al.	
2005/0004665	A1	1/2005	Aklog	
2005/0004668	A1	1/2005	Aklog et al.	
2005/0021056	A1	1/2005	St et al.	
2005/0021057	A1	1/2005	St et al.	
2005/0021058	A1	1/2005	Negro	
2005/0033446	A1	2/2005	Deem et al.	
2005/0038383	A1	2/2005	Kelley et al.	
2005/0038508	A1	2/2005	Gabbay	
2005/0049698	A1	3/2005	Bolling et al.	
2005/0055089	A1	3/2005	Macoviak et al.	
2005/0059351	A1	3/2005	Cauwels et al.	
2005/0065453	A1	3/2005	Shabaz et al.	
2005/0085903	A1	4/2005	Lau	
2005/0119735	A1	6/2005	Spence et al.	
2005/0131438	A1	6/2005	Cohn	
2005/0143809	A1	6/2005	Salahieh et al.	
2005/0149014	A1	7/2005	Hauck et al.	
2005/0159763	A1	7/2005	Mollenauer et al.	
2005/0159810	A1	7/2005	Filsoufi	
2005/0192633	A1	9/2005	Montpetit	
2005/0197694	A1	9/2005	Pai et al.	
2005/0197695	A1	9/2005	Stacchino et al.	
2005/0216039	A1	9/2005	Lederman	
2005/0228422	A1	10/2005	Machold et al.	
2005/0228495	A1	10/2005	Macoviak	
2005/0251001	A1	11/2005	Hassett	
2005/0256452	A1	11/2005	Demarchi et al.	
2005/0267493	A1	12/2005	Schreck et al.	
2005/0273160	A1	12/2005	Lashinski et al.	
2005/0277876	A1	12/2005	Hayden	
2005/0287493	A1	12/2005	Novak et al.	
2006/0004247	A1	1/2006	Kute et al.	
2006/0009759	A1	1/2006	Chrisitian et al.	
2006/0015003	A1	1/2006	Moaddes et al.	
2006/0015179	A1	1/2006	Bulman-Fleming et al.	
2006/0020275	A1	1/2006	Goldfarb et al.	
2006/0020327	A1	1/2006	Lashinski et al.	
2006/0020334	A1*	1/2006	Lashinski	A61F 2/014 623/2.11
2006/0030866	A1	2/2006	Schreck	
2006/0030867	A1	2/2006	Zadno	
2006/0030885	A1	2/2006	Hyde	
2006/0058871	A1	3/2006	Zakay et al.	
2006/0064115	A1	3/2006	Allen et al.	
2006/0064116	A1	3/2006	Allen et al.	
2006/0064118	A1	3/2006	Kimblad	
2006/0074484	A1	4/2006	Huber	
2006/0089671	A1	4/2006	Goldfarb et al.	
2006/0089711	A1	4/2006	Dolan	
2006/0135961	A1	6/2006	Rosenman et al.	
2006/0135993	A1	6/2006	Seguin	
2006/0184198	A1	8/2006	Bales et al.	
2006/0184203	A1	8/2006	Martin et al.	
2006/0195012	A1	8/2006	Mortier et al.	
2006/0229708	A1	10/2006	Powell et al.	
2006/0252984	A1	11/2006	Rahdert et al.	
2006/0276890	A1	12/2006	Solem et al.	
2007/0016225	A1	1/2007	Nakao	
2007/0038293	A1	2/2007	St et al.	
2007/0060997	A1	3/2007	de Boer	
2007/0073185	A1	3/2007	Nakao	
2007/0100356	A1	5/2007	Lucatero et al.	
2007/0118155	A1	5/2007	Goldfarb et al.	
2007/0129737	A1	6/2007	Goldfarb et al.	
2007/0173757	A1	7/2007	Levine et al.	
2007/0197858	A1	8/2007	Goldfarb et al.	
2007/0198082	A1	8/2007	Kapadia et al.	
2007/0260225	A1	11/2007	Sakakine et al.	
2007/0287884	A1	12/2007	Schena	
2008/0009858	A1	1/2008	Rizvi	
2008/0039935	A1	2/2008	Buch et al.	
2008/0045936	A1	2/2008	Vaska et al.	
2008/0051703	A1	2/2008	Thornton et al.	
2008/0051807	A1	2/2008	St et al.	
2008/0097467	A1	4/2008	Gruber et al.	
2008/0097489	A1	4/2008	Goldfarb et al.	
2008/0167714	A1	7/2008	St et al.	

(56) **References Cited**

U.S. PATENT DOCUMENTS

2008/0183194 A1 7/2008 Goldfarb et al.
2008/0188850 A1 8/2008 Mody et al.
2008/0195126 A1 8/2008 Solem
2008/0243249 A1 10/2008 Kohm et al.
2008/0294175 A1 11/2008 Bardsley et al.
2008/0312496 A1 12/2008 Zwolinski
2009/0012538 A1 1/2009 Saliman et al.
2009/0036768 A1 2/2009 Seehusen et al.
2009/0156995 A1 6/2009 Martin et al.
2009/0163934 A1 6/2009 Raschdorf et al.
2009/0177266 A1 7/2009 Powell et al.
2009/0192510 A1 7/2009 Bahney
2009/0198322 A1 8/2009 Deem et al.
2009/0204005 A1 8/2009 Keast et al.
2009/0209955 A1 8/2009 Forster et al.
2009/0209991 A1 8/2009 Hinchliffe et al.
2009/0270858 A1 10/2009 Hauck et al.
2009/0276039 A1 11/2009 Meretei
2009/0281619 A1 11/2009 Le et al.
2009/0326567 A1 12/2009 Goldfarb et al.
2010/0016958 A1 1/2010 St et al.
2010/0022823 A1 1/2010 Goldfarb et al.
2010/0044410 A1 2/2010 Argentine et al.
2010/0121437 A1 5/2010 Subramanian et al.
2010/0152612 A1 6/2010 Headley et al.
2010/0217261 A1 8/2010 Watson
2010/0262231 A1 10/2010 Tuval et al.
2010/0268226 A1 10/2010 Epp et al.
2010/0298929 A1 11/2010 Thornton et al.
2011/0009864 A1 1/2011 Bucciaglia et al.
2011/0184405 A1 7/2011 Mueller
2011/0224710 A1 9/2011 Bleich
2011/0238052 A1 9/2011 Robinson
2011/0282250 A1 11/2011 Fung et al.
2012/0022527 A1 1/2012 Woodruff et al.
2012/0022640 A1 1/2012 Gross et al.
2012/0065464 A1 3/2012 Ellis et al.
2012/0150194 A1 6/2012 Odermatt et al.
2012/0157765 A1* 6/2012 Mitelberg A61B 1/018
606/147
2012/0172915 A1 7/2012 Fifer et al.
2012/0179184 A1 7/2012 Orlov
2012/0265222 A1 10/2012 Gordin et al.
2012/0310330 A1 12/2012 Buchbinder et al.
2012/0316639 A1 12/2012 Kleinschrodt
2012/0330348 A1 12/2012 Strauss et al.
2013/0041314 A1 2/2013 Dillon
2013/0066341 A1 3/2013 Ketai et al.
2013/0066342 A1 3/2013 Dell et al.
2013/0109910 A1 5/2013 Alexander et al.
2013/0172828 A1 7/2013 Kappel et al.
2013/0317515 A1 11/2013 Kuroda et al.
2014/0039511 A1 2/2014 Morris et al.
2014/0135799 A1 5/2014 Henderson
2014/0228871 A1* 8/2014 Cohen A61B 17/32053
606/170
2014/0276913 A1 9/2014 Tah et al.
2014/0309670 A1 10/2014 Bakos et al.
2014/0324164 A1 10/2014 Gross et al.
2014/0350662 A1 11/2014 Vaturi
2014/0358224 A1 12/2014 Tegels et al.
2014/0364866 A1 12/2014 Dryden et al.
2014/0379074 A1 12/2014 Spence et al.
2015/0005704 A1 1/2015 Heisel et al.
2015/0005801 A1 1/2015 Marquis et al.
2015/0051698 A1 2/2015 Ruyra et al.
2015/0094800 A1 4/2015 Chawla
2015/0112430 A1 4/2015 Creaven et al.
2015/0211946 A1 7/2015 Pons et al.
2015/0230947 A1 8/2015 Krieger et al.
2015/0257877 A1 9/2015 Hernandez
2015/0257883 A1* 9/2015 Basude A61B 17/0682
623/2.11
2015/0306806 A1 10/2015 Dando et al.
2015/0313581 A1 11/2015 Wolfe et al.
2016/0015410 A1 1/2016 Asirvatham et al.
2016/0074165 A1 3/2016 Spence et al.
2016/0174979 A1 6/2016 Wei
2016/0317174 A1* 11/2016 Dake A61B 17/320016
2017/0042678 A1 2/2017 Ganesan et al.
2017/0100183 A1 4/2017 Iaizzo et al.
2017/0143330 A1 5/2017 Basude et al.
2017/0202559 A1 7/2017 Taha
2017/0232238 A1 8/2017 Biller et al.
2018/0008268 A1* 1/2018 Khairkhahan A61B 17/10
2018/0028215 A1 2/2018 Cohen
2018/0092661 A1* 4/2018 Prabhu A61B 18/24
2018/0133010 A1 5/2018 Kizuka
2018/0161159 A1 6/2018 Lee et al.
2018/0360457 A1 12/2018 Ellis et al.
2019/0029790 A1 1/2019 Bak-Boychuk et al.
2019/0183571 A1 6/2019 De Marchena
2019/0298517 A1 10/2019 Sanchez et al.
2019/0307458 A1 10/2019 Mathis et al.
2020/0121460 A1 4/2020 Dale et al.
2021/0113232 A1 4/2021 Ortiz et al.
2021/0145574 A1 5/2021 Childs et al.

FOREIGN PATENT DOCUMENTS

CN 103841899 A 6/2014
CN 104244841 A 12/2014
DE 3504292 C1 7/1986
DE 9100873 U1 4/1991
DE 10116168 A1 11/2001
EP 0179562 A1 4/1986
EP 0558031 A2 9/1993
EP 0684012 A2 11/1995
EP 0727239 A2 8/1996
EP 0782836 A1 7/1997
EP 1230899 A1 8/2002
EP 1674040 A2 6/2006
EP 1980288 A1 10/2008
EP 2005912 A2 12/2008
EP 2537487 A1 12/2012
EP 2641570 A1 9/2013
EP 2702965 A1 3/2014
EP 2740419 A1 6/2014
EP 3009103 A1 4/2016
FR 2705556 A1 12/1994
FR 2768324 A1 3/1999
FR 2903292 A1 1/2008
GB 1598111 A 9/1981
GB 2151142 A 7/1985
JP 09-253030 A 9/1997
JP 11-089937 A 4/1999
JP 2000-283130 A 10/2000
JP 2001-517529 A 10/2001
JP 2006-528911 A 12/2006
JP 2013-516244 A 5/2013
JP 2013-523384 A 6/2013
JP 2014-523274 A 9/2014
JP 2015-502548 A 1/2015
JP 2018-030008 A 3/2018
WO 81/00668 A1 3/1981
WO 91/01689 A1 2/1991
WO 91/18881 A1 12/1991
WO 92/12690 A1 8/1992
WO 94/18881 A1 9/1994
WO 94/18893 A1 9/1994
WO 95/08292 A1 3/1995
WO 95/11620 A2 5/1995
WO 95/15715 A1 6/1995
WO 96/14032 A1 5/1996
WO 96/20655 A1 7/1996
WO 96/22735 A1 8/1996
WO 96/30072 A1 10/1996
WO 97/18746 A2 5/1997
WO 97/25927 A1 7/1997
WO 97/26034 A1 7/1997
WO 97/38748 A2 10/1997
WO 97/39688 A2 10/1997
WO 97/48436 A2 12/1997
WO 98/07375 A1 2/1998

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

WO	98/24372	A1	6/1998
WO	98/30153	A1	7/1998
WO	98/32382	A1	7/1998
WO	98/35638	A1	8/1998
WO	99/00059	A1	1/1999
WO	99/01377	A1	1/1999
WO	99/07295	A1	2/1999
WO	99/07354	A2	2/1999
WO	99/13777	A1	3/1999
WO	99/44524	A2	9/1999
WO	99/66967	A1	12/1999
WO	00/02489	A1	1/2000
WO	00/03651	A1	1/2000
WO	00/03759	A2	1/2000
WO	00/12168	A1	3/2000
WO	00/44313	A1	8/2000
WO	00/59382	A1	10/2000
WO	00/60995	A2	10/2000
WO	01/00111	A1	1/2001
WO	01/00114	A1	1/2001
WO	01/03651	A2	1/2001
WO	01/26557	A1	4/2001
WO	01/26586	A1	4/2001
WO	01/26587	A1	4/2001
WO	01/26588	A2	4/2001
WO	01/26703	A1	4/2001
WO	01/28432	A1	4/2001
WO	01/28455	A1	4/2001
WO	01/47438	A1	7/2001
WO	01/49213	A2	7/2001
WO	01/50985	A1	7/2001
WO	01/54618	A1	8/2001
WO	01/56512	A1	8/2001
WO	01/66001	A2	9/2001
WO	01/70320	A1	9/2001
WO	01/89440	A2	11/2001
WO	01/95831	A2	12/2001
WO	01/95832	A2	12/2001
WO	01/97741	A2	12/2001
WO	02/00099	A2	1/2002
WO	02/01999	A2	1/2002
WO	02/03892	A1	1/2002
WO	02/34167	A2	5/2002
WO	02/60352		8/2002
WO	02/62263		8/2002
WO	02/62270		8/2002
WO	02/62408		8/2002
WO	03/01893	A2	1/2003
WO	03/03930		1/2003
WO	03/20179		3/2003
WO	03/28558	A2	4/2003
WO	03/37171		5/2003
WO	03/47467		6/2003
WO	03/49619		6/2003
WO	03/73910		9/2003
WO	03/73913		9/2003
WO	03/82129		10/2003
WO	2003/105667		12/2003
WO	2004/004607	A1	1/2004
WO	2004/006810	A1	1/2004
WO	2004/012583	A2	2/2004
WO	2004/012789	A2	2/2004
WO	2004/014282	A2	2/2004
WO	2004/019811	A2	3/2004
WO	2004/030570	A2	4/2004
WO	2004/037317	A2	5/2004
WO	2004/045370	A2	6/2004
WO	2004/045378	A2	6/2004
WO	2004/045463	A2	6/2004
WO	2004/047679	A1	6/2004
WO	2004/062725	A1	7/2004
WO	2004/082523	A2	9/2004
WO	2004/082538	A2	9/2004
WO	2004/093730	A2	11/2004
WO	2004/103162	A2	12/2004
WO	2004/112585	A2	12/2004
WO	2004/112651	A2	12/2004
WO	2005/002424	A2	1/2005
WO	2005/018507	A2	3/2005
WO	2005/027797	A1	3/2005
WO	2005/032421	A2	4/2005
WO	2005/062931	A2	7/2005
WO	2005/112792	A2	12/2005
WO	2006/037073	A2	4/2006
WO	2006/105008	A1	10/2006
WO	2006/105009	A1	10/2006
WO	2006/113906	A1	10/2006
WO	2006/115875	A2	11/2006
WO	2006/115876	A2	11/2006
WO	2007/136829	A1	11/2007
WO	2008/103722	A2	8/2008
WO	2010/024801	A1	3/2010
WO	2010/121076	A2	10/2010
WO	2012/020521	A1	2/2012
WO	2013/049734	A1	4/2013
WO	2013/103934	A1	7/2013
WO	2014/064694	A2	5/2014
WO	2014/121280	A2	8/2014
WO	2016/022797	A1	2/2016
WO	2016/144708	A1	9/2016
WO	2016/150806	A1	9/2016
WO	2017/223073	A1	12/2017
WO	2018/009718	A1	1/2018
WO	2018/106482	A1	6/2018
WO	2018/236766	A1	12/2018
WO	2019/040943	A1	2/2019
WO	2019/195336	A1	10/2019

OTHER PUBLICATIONS

Langer et al., "Posterier mitral leaflet extensions: An adjunctive repair option for ischemic mitral regurgitation?" J Thorac Cardiovasc Surg, 131:868-877 (2006).

Lorusso et al., "The double-orifice technique for mitral valve reconstruction: predictors of postoperative outcome," Eur J. Cardiothorac Surg, 20:583-589 (2001).

Maisano et al, The Edge-to-edge Technique: A Simplified Method to Correct Mitral Insufficiency, Eur. J. Cardiothorac. Surg., Jan. 14, 1998, pp. 240-246, vol. 13.

Maisano et al, The future of transcatheter mitral valve interventions: competitive or complementary role of repair vs. replacement? Eur Heart J. Jul. 7, 2015; 36(26):1651-1659.

Maisano et al., "The double orifice repair for Barlow Disease: a simple solution for a complex repair," Supplement I Circulation, (Nov. 1999); 100(18):1-94.

Maisano et al., "The double orifice technique as a standardized approach to treat mitral regurgitation due to severe myxomatous disease: surgical technique," European Journal of Cardio-thoracic Surgery, 17:201-205 (2000).

Maisano et al., "The Future of Transcatheter Mitral Valve Interventions: Competitive or Complementary Role of Repair vs. Replacement? ", Eur Heart J.36(26):1651-1659 (Jul. 7, 2015).

Maisano et al., "The hemodynamic effects of double-orifice valve repair for mitral regurgitation: a 3D computational model," European Journal of Cardio-thoracic Surgery, 15:419-425 (1999).

Maisano et al., "Valve repair for traumatic tricuspid regurgitation," Eur. J. Cardio-thorac Surg, (1996) 10:867-873.

Mantovani et al., "Edge-to-edge Repair of Congenital Familiar Tricuspid Regurgitation: Case Report," J. Heart Valve Dis., 9:641-643 (2000).

McCarthy et al, "Tricuspid Valve Repair With the Cosgrove-Edwards Annuloplasty System", Ann. Thorac. Surg., 64:267-8 (Jan. 16, 1997).

McCarthy et al., "Partial left ventriculectomy and mitral valve repair for end-stage congestive heart failure," European Journal of Cardio-thoracic Surgery, 13:337-343 (1998).

McCarthy et al., "Tricuspid Valve Repair With the Cosgrove-Edwards Annuloplasty System", Ann. Throac Surg. 64:267-8 (Jan. 16, 1997).

(56) **References Cited**

OTHER PUBLICATIONS

Moainie et al., "Correction of Traumatic Tricuspid Regurgitation Using the Double Orifice Technique," Annals of Thoracic Surgery, 73:963-965 (2002).
Morales et al., "Development of an Off Bypass Mitral Valve Repair," The Heart Surgery Forum #1999-4693, 2(2):115-120 (1999).
Nakanishi et al., "Early Outcome with the Alfieri Mitral Valve Repair," J. Cardiol., 37: 263-266 (2001) [Abstract in English; Article in Japanese].
Nielsen et al., "Edge-to-Edge Mitral Repair: Tension of the Approximating Suture and Leaflet Deformation During Acute Ischemic Mitral Regurgitation in the Ovine Heart," Circulation, 104(Suppl. 1):1-29-1-35 (2001).
Nishimura, et al. 2014 AHA/ACC guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. J Am Coll Cardiol. Jun. 10, 2014;63(22):2438-88.
Noera et al., "Tricuspid Valve Incompetence Caused by Nonpenetrating Thoracic Trauma", Annals of Thoracic Surgery, 51:320-322 (1991).
Notice of Allowance received for U.S. Appl. No. 14/216,787, filed Nov. 7, 2016.
Notice of Allowance received for U.S. Appl. No. 14/216,787, mailed on Nov. 7, 2016.
Notice of Allowance received for U.S. Appl. No. 14/577,852, filed Apr. 25, 2018.
Notice of Allowance received for U.S. Appl. No. 14/577,852, mailed on Apr. 25, 2018.
Notice of Allowance received for U.S. Appl. No. 15/642,245, mailed on Jan. 29, 2020.
Notice of Allowance received for U.S. Appl. No. 15/642,245, mailed on Mar. 27, 2020.
Notice of Allowance received for U.S. Appl. No. 15/642,245, mailed on Nov. 6, 2019.
Notice of Allowance received for U.S. Appl. No. 15/423,060, mailed on Jan. 27, 2020.
Office Action received for U.S. Appl. No. 14/216,787, filed Apr. 8, 2016.
Office Action received for U.S. Appl. No. 14/216,787, mailed on Apr. 8, 2016.
Office Action received for U.S. Appl. No. 14/216,813, filed Apr. 6, 2018.
Office Action received for U.S. Appl. No. 14/216,813, filed Dec. 15, 2017.
Office Action received for U.S. Appl. No. 14/216,813, filed Mar. 9, 2017.
Office Action received for U.S. Appl. No. 14/216,813, mailed on Apr. 6, 2018.
Office Action received for U.S. Appl. No. 14/216,813, mailed on Dec. 15, 2017.
Office Action received for U.S. Appl. No. 14/216,813, mailed on Mar. 9, 2017.
Office Action received for U.S. Appl. No. 14/577,852, filed May 16, 2017.
Office Action received for U.S. Appl. No. 14/577,852, filed Oct. 20, 2016.
Office Action received for U.S. Appl. No. 14/577,852, filed Sep. 7, 2017.
Office Action received for U.S. Appl. No. 14/577,852, mailed on May 16, 2017.
Office Action received for U.S. Appl. No. 14/577,852, mailed on Oct. 20, 2016.
Office Action received for U.S. Appl. No. 14/577,852, mailed on Sep. 7, 2017.
Office Action received for U.S. Appl. No. 15/423,060, mailed on Apr. 25, 2019.
Office Action received for U.S. Appl. No. 15/423,060, mailed on Aug. 19, 2019.
Office Action received for U.S. Appl. No. 15/423,060, mailed on Oct. 28, 2019.
Office Action received for U.S. Appl. No. 15/642,245, mailed on Aug. 9, 2019.
Office Action received for U.S. Appl. No. 15/724,545, filed Dec. 27, 2019.
Office Action received for U.S. Appl. No. 15/724,545, mailed on Dec. 27, 2019.
Office Action received for U.S. Appl. No. 15/724,545, mailed on May 1, 2020.
Osawa et al., "Partial Left Ventriculectomy in a 3-Year Old Boy with Dilated Cardiomyopathy," Japanese Journal of Thoracic and Cardiovascular Surg, 48:590-593 (2000).
Park et al, Clinical Use of Blade Atrial Septostomy, Circulation, 1978, pp. 600-608, vol. 58.
Patel et al., #57 Epicardial Atrial Defibrillation: Novel Treatment of Postoperative Atrial Fibrillation, 2003 STS Presentation, [Abstract Only].
Privitera et al., "Alfieri Mitral Valve Repair: Clinical Outcome and Pathology," Circulation, 106:e173-e174 (2002).
Redaelli et al., "A Computational Study of the Hemodynamics After 'Edge-To-Edge' Mitral Valve Repair," Journal of Biomechanical Engineering, 123:565-570 (2001).
Reul et al., "Mitral Valve Reconstruction for Mitral Insufficiency," Progress in Cardiovascular Diseases, XXXIX(6):567-599 (1997).
Ricchi et al, Linear Segmental Annuloplasty for Mitral Valve Repair, Ann. Thorac. Surg., Jan. 7, 1997, pp. 1805-1806, vol. 63.
Robicsek et al., #60 The Bicuspid Aortic Valve: How Does It Function? Why Does It Fail? 2003 STS Presentation, [Abstract Only].
Rose et al., "Late MitraClip Failure: Removal Technique for Leaflet-Sparing Mitral Valve Repair", Journal of Cardiac Surgery, (Jul. 4, 2012), XP055047339, DOI: 10.1111/j. 1540-8191.2012.01483.x [retrieved on Dec. 11, 2012].
Supplemental European Search Report of EP Application No. 02746781, dated May 13, 2008, 3 pages total.
Supplementary European Search Report issued in European Application No. 05753261.6 dated Jun. 9, 2011, 3 pages total.
Tager et al, Long-Term Follow-Up of Rheumatic Patients Undergoing Left-Sided Valve Replacement With Tricuspid Annuloplasty—Validity of Preoperative Echocardiographic Criteria in the Decision to Perform Tricuspid Annuloplasty, Am. J. Cardiol., Apr. 15, 1998, pp. 1013-1016, vol. 81.
Takizawa H et al: Development of a microfine active bending catheter equipped with MIF tactile sensors", Micro Electro Mechanical Systems, 1999. MEMS '99. Twelfth IEEE Interna Tional Conference on Orlando, FL, USA Jan. 17-21, 1999, Piscataway, NJ, USA,IEEE, US, Jan. 17, 1999 (Jan. 17, 1999), pp. 412-417, XP010321677, ISBN: 978-0-7803-5194-3 figures 1-3.".
Tamura et al., "Edge to Edge Repair for Mitral Regurgitation in a Patient with Chronic Hemodialysis: Report of a Case," Kyobu Geka. The Japanese Journal of Thoracic Surgery, 54(9):788-790 (2001).
Tibayan et al., #59 Annular Geometric Remodeling in Chronic Ischemic Mitral Regurgitation, 2003 STS Presentation, [Abstract Only].
Timek et al., "Edge-to-edge mitral repair: gradients and three-dimensional annular dynamics in vivo during inotropic stimulation," Eur J. of Cardiothoracic Surg., 19:431-437 (2001).
Timek, "Edge-to-Edge Mitral Valve Repair without Annuloplasty Ring in Acute Ischemic Mitral Regurgitation," [Abstract] Clinical Science, Abstracts from Scientific Sessions, 106(19):2281 (2002).
Totaro, "Mitral valve repair for isolated prolapse of the anterior leaflet: an 11-year follow-up," European Journal of Cardio-thoracic Surgery, 15:119-126 (1999).
U.S. Provisional Application Filed on Jul. 6, 2016, by Khairkhahan., U.S. Appl. No. 62/359,121.
U.S. Provisional Application Filed on Nov. 7, 2016, by Khairkhahan., U.S. Appl. No. 62/418,571.
U.S. Provisional Application Filed on Oct. 22, 2018, by Dale et al., U.S. Appl. No. 62/748,947.

(56) **References Cited**

OTHER PUBLICATIONS

Uchida et al., Percutaneous Cardiomyotomy and Valvulotomy with Angioscopic Guidance, Am. Heart J., Apr. 1991, pp. 1221-1224, vol. 121.
Umana et al, 'Bow-Tie' Mitral Valve Repair: An Adjuvant Technique for Ischemic Mitral Regurgitation, Ann. Thorac. Surg., May 12, 1998, pp. 1640-1646, vol. 66.
Umana et al., "'Bow-tie' Mitral Valve Repair Successfully Addresses Subvalvular Dysfunction in Ischemic Mitral Regurgitation," Surgical Forum, XLVIII:279-280 (1997).
U.S. Appl. No. 14/216,813, filed Mar. 17, 2014, Hernandez.
Votta et al., "3-D Computational Analysis of the Stress Distribution on the Leaflets after Edge-to-Edge Repair of Mitral Regurgitation," Journal of Heart Valve Disease, 11:810-822 (2002).
Abe et al, "De Vega's Annuloplasty for Acquired Tricuspid Disease: Early and Late Results in 110 Patients", Ann. Thorac. Surg., pp. 670-676, vol. 48 (Jan. 1989).
Abe et al., "Updated in 1996—De Vega's Annuloplasty for Acquired Tricuspid Disease: Early and Late Results in 110 Patients", Ann. Thorac. Surg., pp. 1876-1877, vol. 62 (1996).
Agricola et al., "Mitral Valve Reserve in Double Orifice Technique: an Exercise Echocardiographic Study," Journal of Heart Valve Disease, 11(5):637-643 (2002).
Alfieri et al., "An Effective Technique to Correct Anterior Mitral Leaflet Prolapse," J. Card Surg., 14:468-470 (1999).
Alfieri et al., "Novel Suture Device for Beating Heart Mitral Leaflet Approximation," Annals of Thoracic Surgery, 74:1488-1493 (2002).
Alfieri et al., "The double orifice technique in mitral valve repair: a simple solution for complex problems," Journal of Thoracic and Cardiovascular Surgery, 122:674-681 (2001).
Alfieri et al., "The Edge to Edge Technique," The European Association For Cardio-Thoracic Surgery, 14th Annual Meeting, Frankfurt/ Germany, Oct. 7-11, 2000, Post Graduate Courses, Book of Proceedings.
Alfieri, "The Edge-to-Edge Repair of the Mitral Valve," [Abstract] 6th Annual New Era Cardiac Care: Innovation & Technology, Heart Surgery Forum, (Jan. 2003) pp. 103.
Ali Khan et al, Blade Atrial Septostomy: Experience with the First 50 Procedures, Cathet. Cardiovasc. Diagn., Aug. 1991, pp. 257-262, vol. 23.
Alvarez et al, Repairing the Degenerative Mitral Valve: Ten to Fifteen-year Follow-up, Journal Thoracic of Cardiovascular Surgery, Aug. 1996, pp. 238-247, vol. 112, No. 2.
Arisi et al., "Mitral Valve Repair with Alfieri Technique in Mitral Regurgitation of Diverse Etiology: Early Echocardiographic Results," Circulation Supplement II, 104(17):3240 (2001).
Bach et al, Early Improvement in Congestive Heart Failure After Correction of Secondary Mitral Regurgitation in End-stage Cardiomyopathy, American Heart Journal, Jun. 1995, pp. 1165-1170, vol. 129, No. 6.
Bach et al, Improvement Following Correction of Secondary Mitral Regurgitation in End-stage Cardiomyopathy With Mitral Annuloplasty, Am. J. Cardiol., Oct. 15, 1996, pp. 966-969, vol. 78.
Bailey, "Mitral Regurgitation" in Surgery of the Heart, Chapter 20, pp. 686-737 (1955).
Bernal et al., "The Valve Racket: a new and different concept of atrioventricular valve repair," Eur. J. Cardio-thoracic Surgery 29:1026-1029 (2006).
Bhudia et al., "Edge-to-Edge (Alfieri) Mitral Repair: Results in Diverse Clinical Settings," Ann Thorac Surg, 77:1598-1606 (2004).
Bhudia, #58 Edge-to-edge mitral repair: a versatile mitral repair technique, 2003 STS Presentation, [Abstract Only], 2004.
Bolling et al, Surgery for Acquired Heart Disease: Early Outcome of Mitral Valve Reconstruction in Patients with End-stage Cardiomyopathy, Journal of Thoracic and Cariovascular Surgery, Apr. 1995, pp. 676-683, vol. 109, No. 4.
Borghetti et al., "Preliminary observations on haemodynamics during physiological stress conditions following 'double-orifice' mitral valve repair," European Journal of Cardio-thoracic Surgery, 20:262-269 (2001).
Castedo, "Edge-to-Edge Tricuspid Repair for Redeveloped Valve Incompetence after DeVega's Annuloplasty," Ann Thora Surg., 75:605-606 (2003).
Chinese Office Action issued in Chinese Application No. 200980158707.2 dated Sep. 9, 2013.
Communication dated Apr. 16, 2018 from the European Patent Office in counterpart European application No. 04752603.3.
Communication dated Apr. 28, 2017 issued by the European Patent Office in counterpart application No. 16196023.2.
Communication dated Jan. 26, 2017, from the European Patent Office in counterpart European application No. 16196023.2.
Communication dated May 8, 2017, from the European Patent Office in counterpart European Application No. 04752714.8.
Dang N C et al., "Surgical Revision After Percutaneous Mitral Valve Repair with a Clip: Initial Multicenter Experience", The Annals of Thracic Surgery, Elsevier, United States, vol. 80, No. 6, pp. 2338-2342, (Dec. 1, 2005), XP027732951, ISSN:0003-4975 [retrieved on Dec. 1, 2005].
Dec et al, Idiopathic Dilated Cardiomyopathy, The New England Journal of Medicine, Dec. 8, 1994, pp. 1564-1575, vol. 331, No. 23.
Dottori et al., "Echocardiographic imaging of the Alfieri type mitral valve repair," Ital. Heart J., 2(4):319-320 (2001).
Downing et al., "Beating heart mitral valve surgery: Preliminary model and methodology," Journal of Thoracic and Cardiovascular Surgery, 123(6):1141-1146 (2002).
Extended European Search Report, dated Oct. 17, 2014, issued in European Patent Application No. 06751584.1.
Falk et al., "Computer-Enhanced Mitral Valve Surgery: Toward a Total Endoscopic Procedure," Seminars in Thoracic and Cardiovascular Surgery, 11(3):244-249 (1999).
Feldman, et al. Randomized Comparison of Percutaneous Repair and Surgery for Mitral Regurgitation: 5-Year Results of Everest II. J Am Coll Cardiol. Dec. 29, 2015;66(25):2844- 2854.
Filsoufi et al., "Restoring Optimal Surface of Coaptation With a Mini Leaflet Prosthesis: A New Surgical Concept for the Correction of Mitral Valve Prolapse," Intl. Soc. for Minimally Invasive Cardiothoracic Surgery 1(4):186-87 (2006).
Frazier et al., #62 Early Clinical Experience with an Implantable, Intracardiac Circulatory Support Device: Operative Considerations and Physiologic Implications, 2003 STS Presentation, 1 page total. [Abstract Only].
Fucci et al, Improved Results with Mitral Valve Repair Using New Surgical Techniques, Eur. J. Cardiothorac. Surg., Nov. 1995, pp. 621-627, vol. 9.
Fundaro et al., "Chordal Plication and Free Edge Remodeling for Mitral Anterior Leaflet Prolapse Repair: 8-Year Follow-up," Annals of Thoracic Surgery, 72:1515-1519 (2001).
Garcia-Rinaldi et al., "Left Ventricular Volume Reduction and Reconstruction is Ischemic Cardiomyopathy," Journal of Cardiac Surgery, 14:199-210 (1999).
Gateliene, "Early and postoperative results results of metal and tricuspid valve insufficiency surgical treatment using edge-to-edge central coaptation procedure," (Oct. 2002) 38 (Suppl 2):172 175.
Gatti et al., "The edge to edge technique as a trick to rescue an imperfect mitral valve repair," Eur. J. Cardiothorac Surg, 22:817-820 (2002).
Gundry, "Facile mitral valve repair utilizing leaflet edge approximation: midterm results of the Alfieri figure of eight repair," Presented at the Meeting of the Western Thoracic Surgical Association, (1999).
Gupta et al., #61 Influence of Older Donor Grafts on Heart Transplant Survival: Lack of Recipient Effects, 2003 STS Presentation, [Abstract Only].
Ikeda et al., "Batista's Operation with Coronary Artery Bypass Grafting and Mitral Valve Plasty for Ischemic Dilated Cardiomyopathy," The Japanese Journal of Thoracic and Cardiovascular Surgery, 48:746-749 (2000).
Izzat et al., "Early Experience with Partial Left Ventriculectomy in the Asia-Pacific Region," Annuals of Thoracic Surgery, 67:1703-1707 (1999).
Kallner et al., "Transaortic Approach for the Alfieri Stitch," Ann Thorac Surg, 71:378-380 (2001).

(56) **References Cited**

OTHER PUBLICATIONS

Kameda et al, Annuloplasty for Severe Mitral Regurgitation Due to Dilated Cardiomyopathy, Ann. Thorac. Surg., 1996, pp. 1829-1832, vol. 61.
Kavarana et al., "Transaortic Repair of Mitral Regurgitation," The Heart Surgery Forum, #2000-2389, 3(1):24-28 (2000).
Kaza et al., "Ventricular Reconstruction Results in Improved Left Ventricular Function and Amelioration of Mitral Insufficiency," Annals of Surgery, 235(6):828-832 (2002).
Khan et al., "Blade Atrial Septostomy; Experience with the First 50 Procedures", Catheterization and Cardiovascular Diagnosis, 23:257-262 (1991).
Kherani et al., "The Edge-To-Edge Mitral Valve Repair: The Columbia Presbyterian Experience," Ann. Thorac. Surg., 78:73-76 (2004).
Kron et al., "Surgical Relocation of the Posterior Papillary Muscle in Chronic Ischemic Mitral Regurgitation," Annals. Of Thoracic Surgery, 74:600-601 (2002).

* cited by examiner

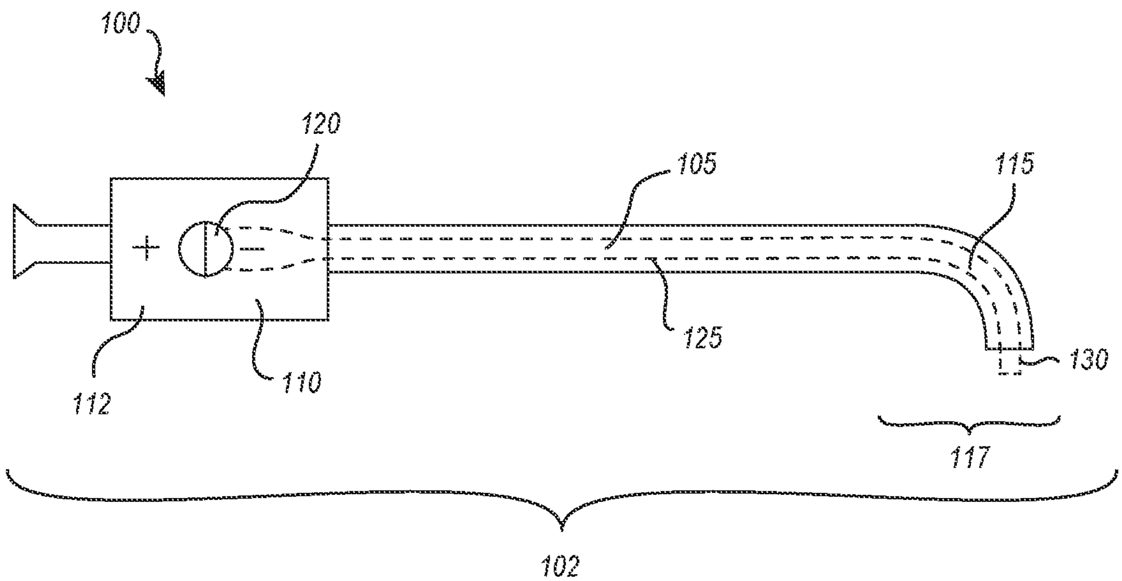


FIG. 1

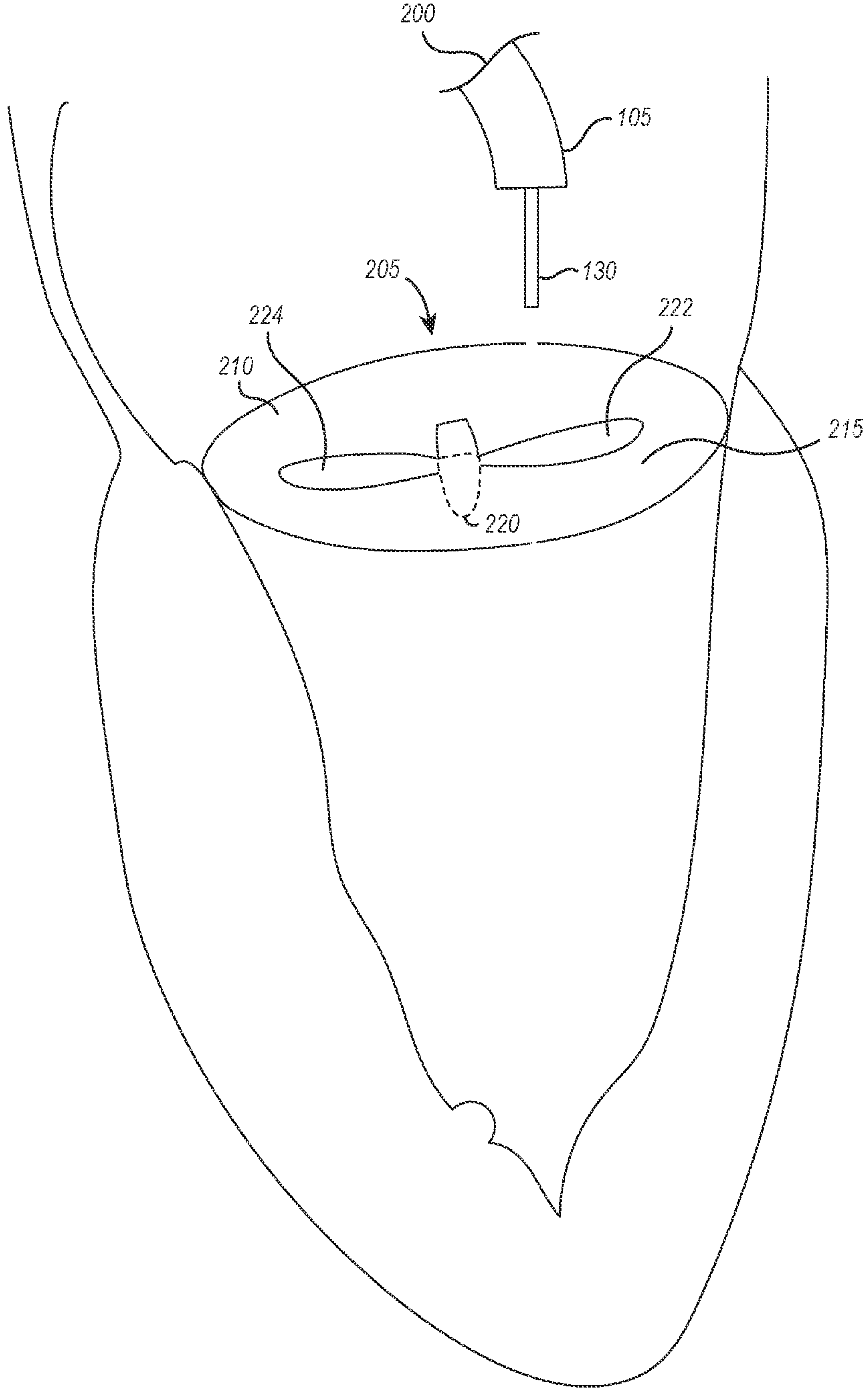


FIG. 2

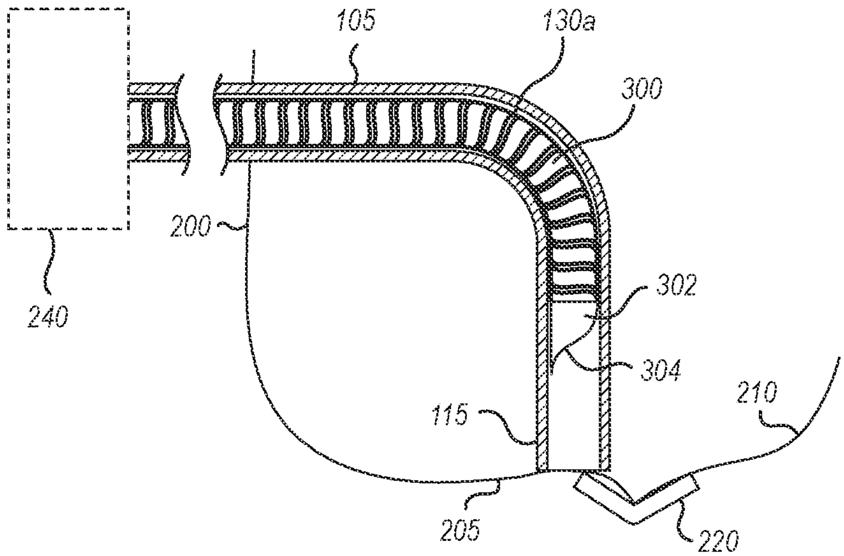

105
130a
300
200
240
302
304
210
115
205
220

FIG. 3A

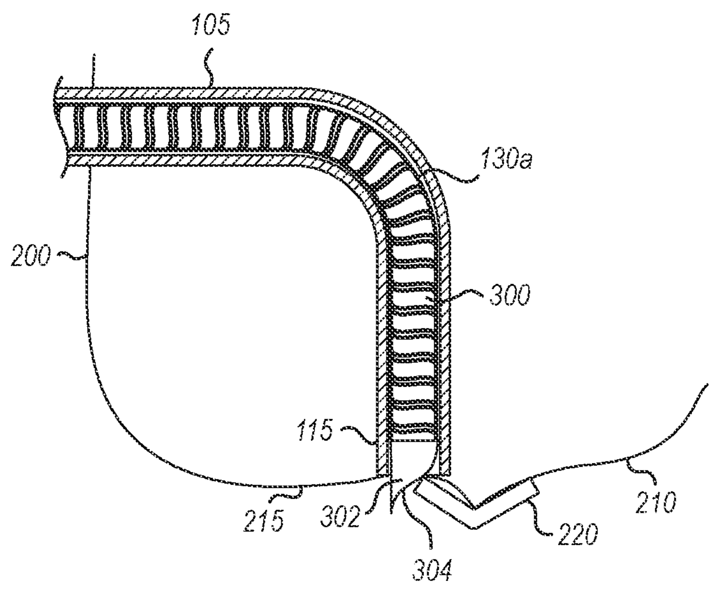

105
130a
200
300
115
210
215
302
220
304

FIG. 3B

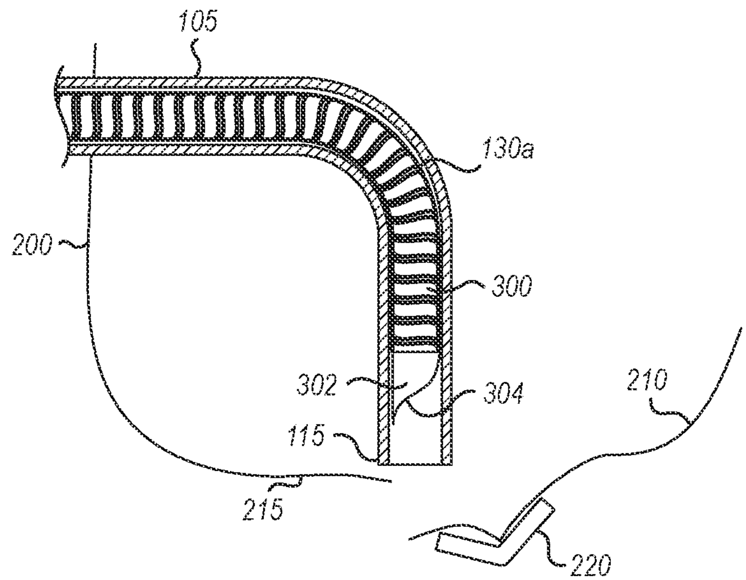


FIG. 3C

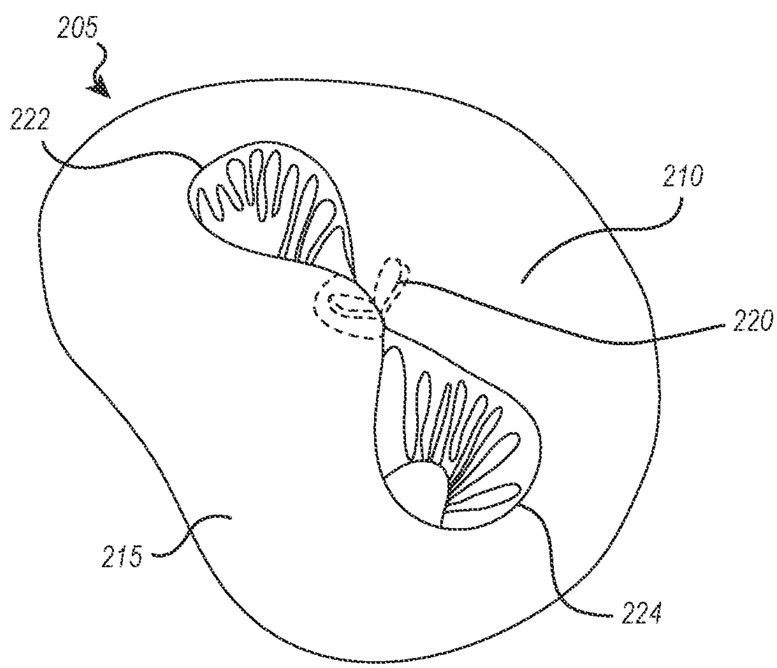


FIG. 3D

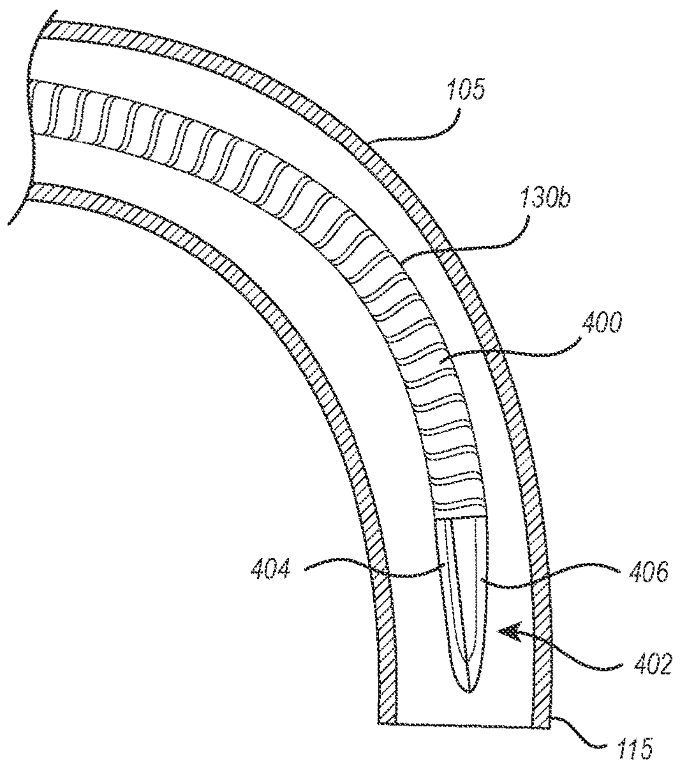

105
130b
400
404
406
402
115

FIG. 4A

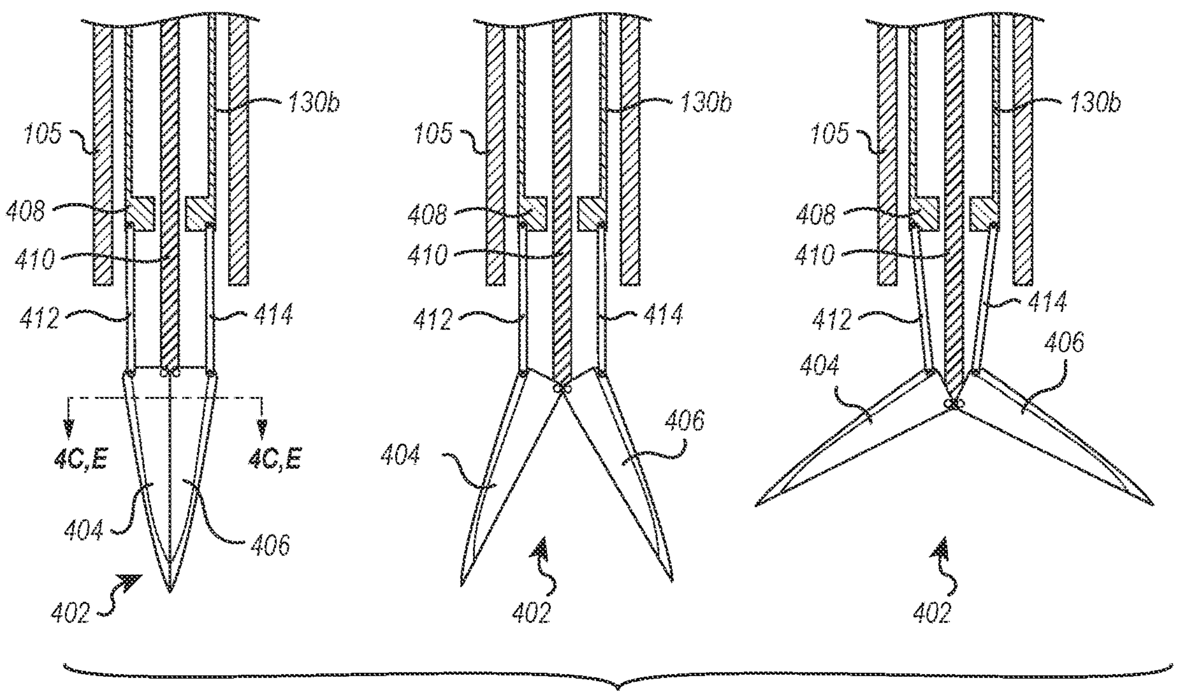

105
130b
408
410
412
414
4C,E
4C,E
404
406
402
105
130b
408
410
412
414
404
406
402
105
130b
408
410
412
414
404
406
402

FIG. 4B

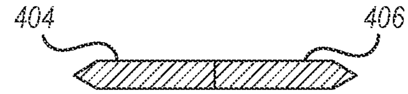


FIG. 4C

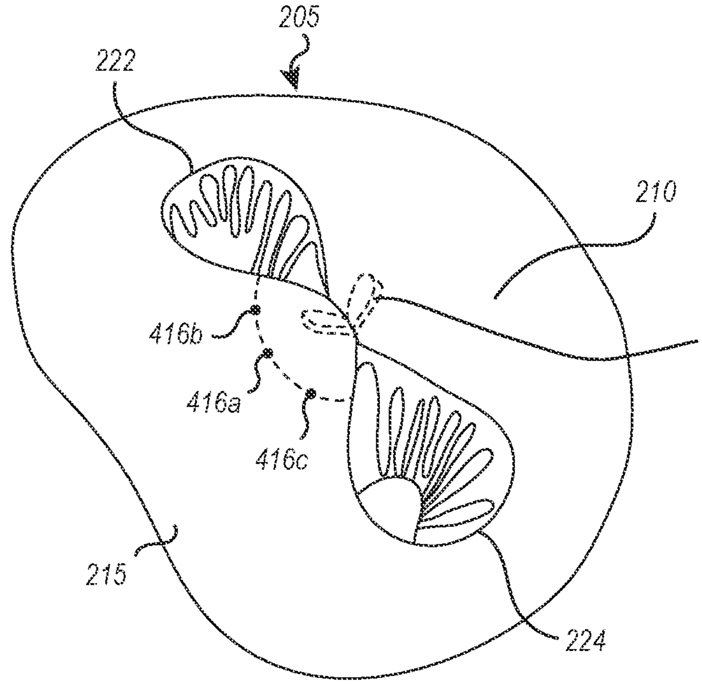


FIG. 4D

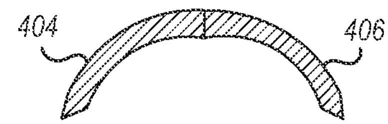


FIG. 4E

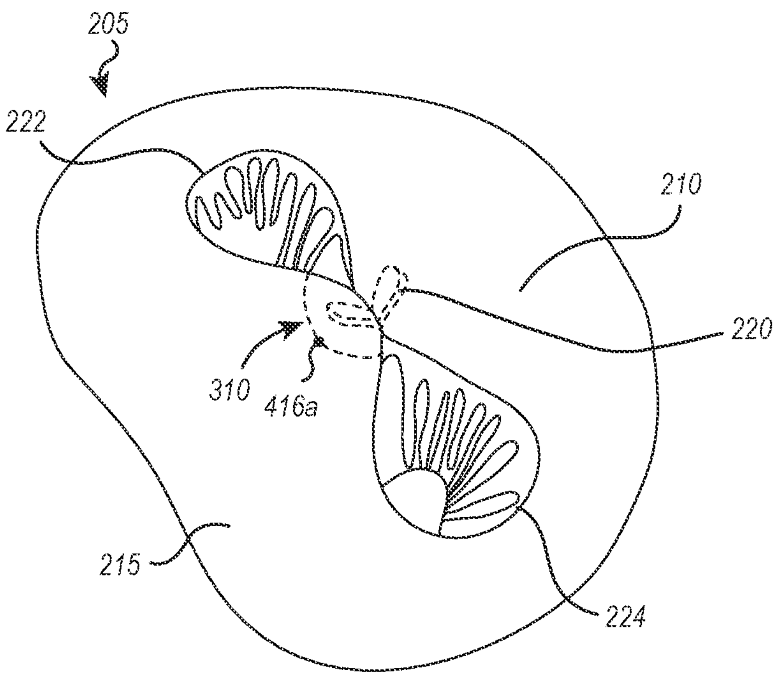


FIG. 4F

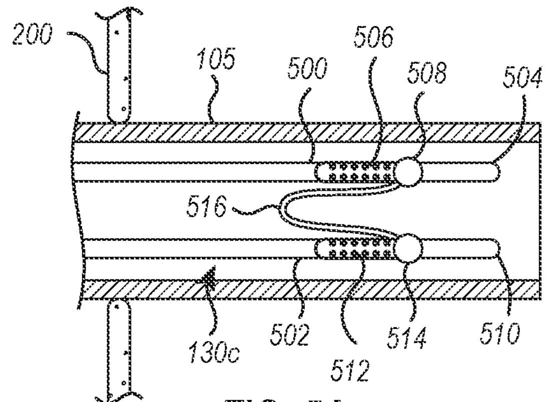

200
105
500
506
508
504
516
130c
502
512
514
510

FIG. 5A

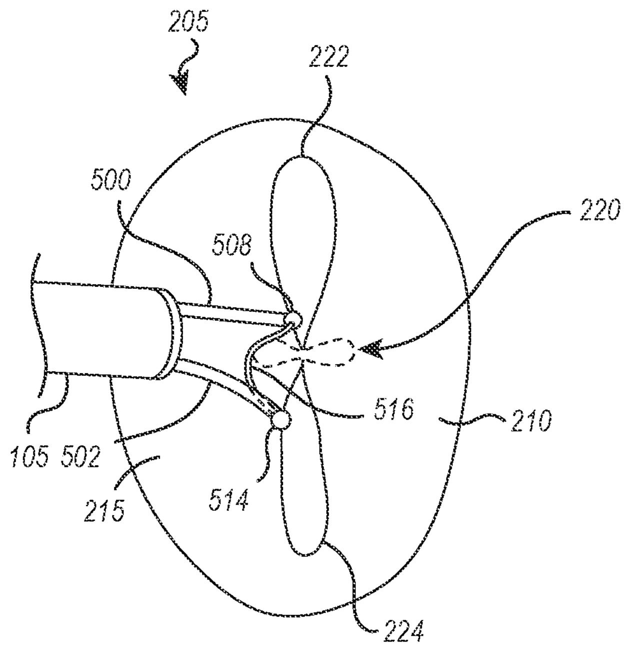

205
222
220
500
508
516
210
105
502
514
215
224

FIG. 5B

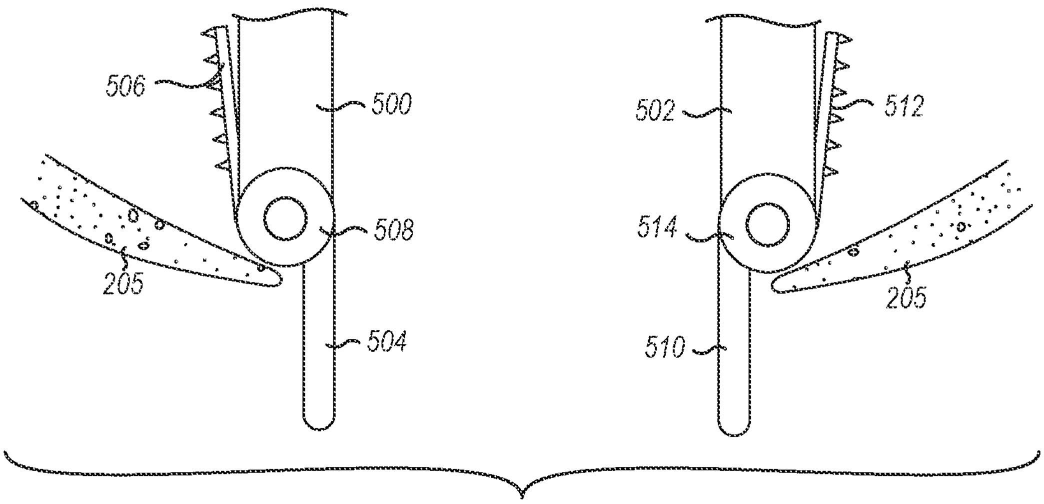


FIG. 5C

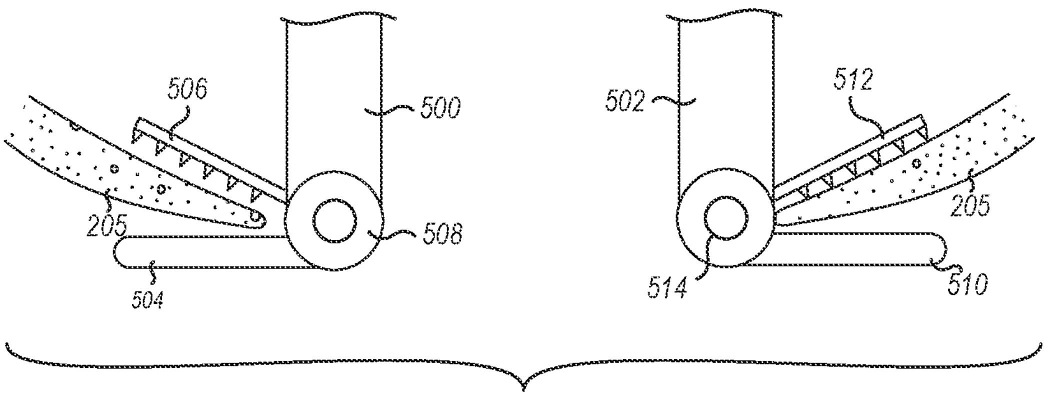


FIG. 5D

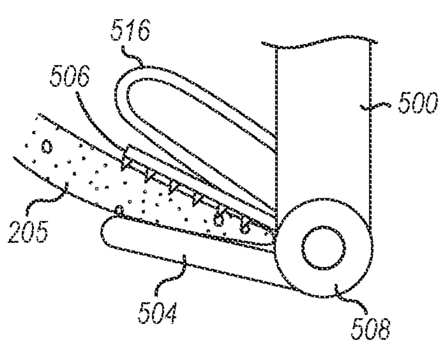


FIG. 5E

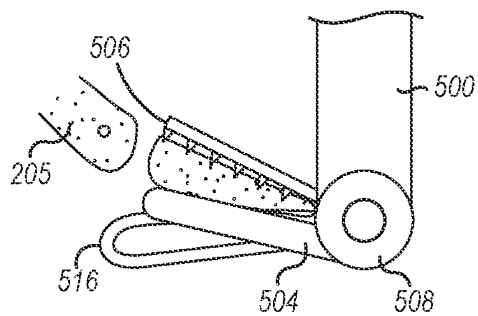


FIG. 5F

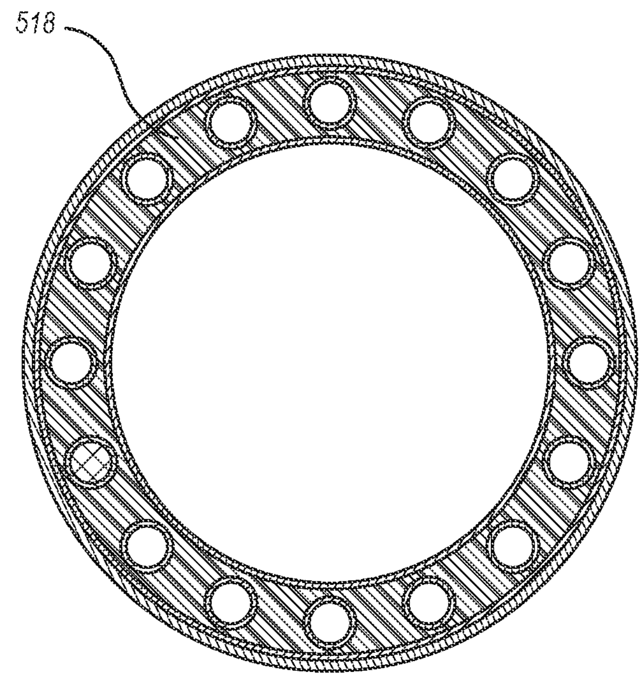


FIG. 5G

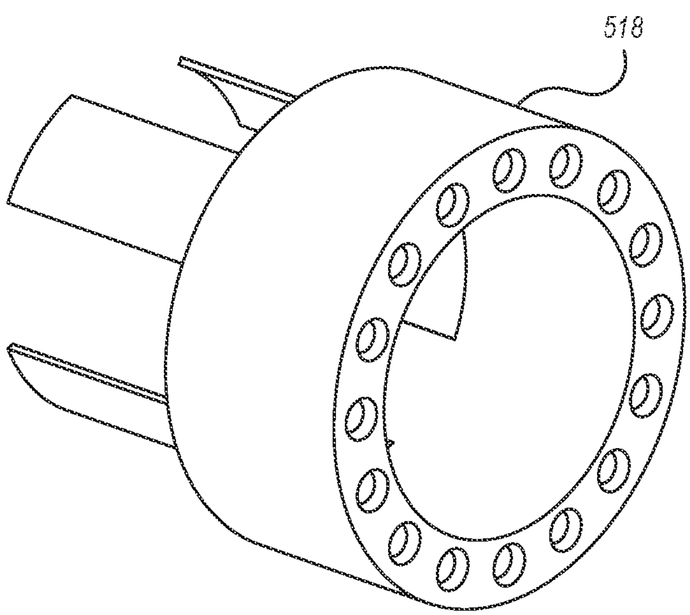


FIG. 5H

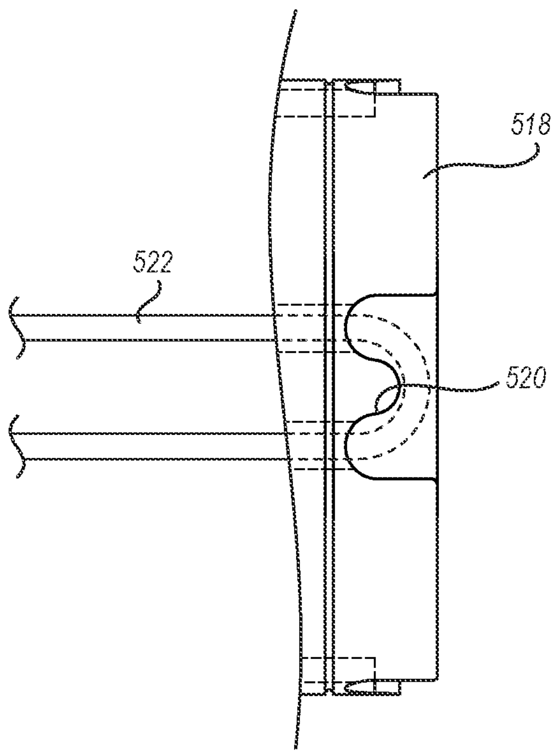


FIG. 5I

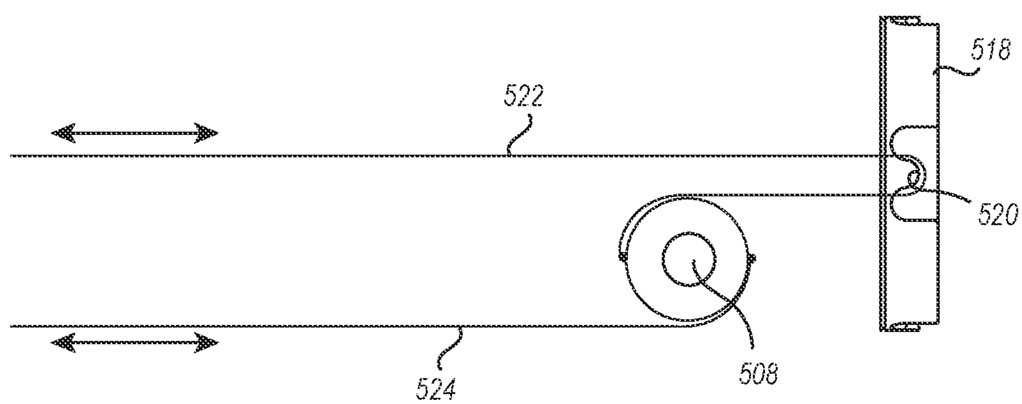


FIG. 5J

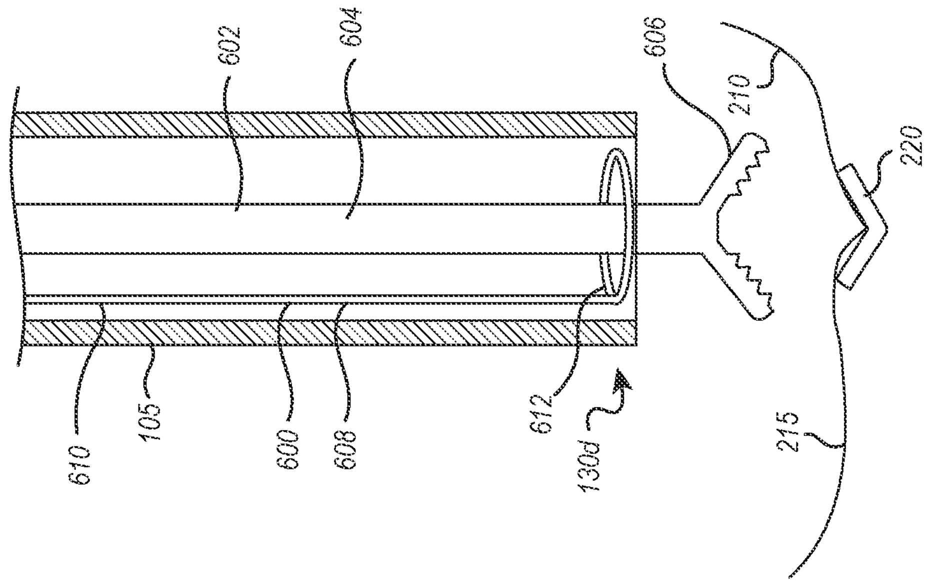

602
604
606
210
220
610
105
600
608
612
130d
215

FIG. 6B

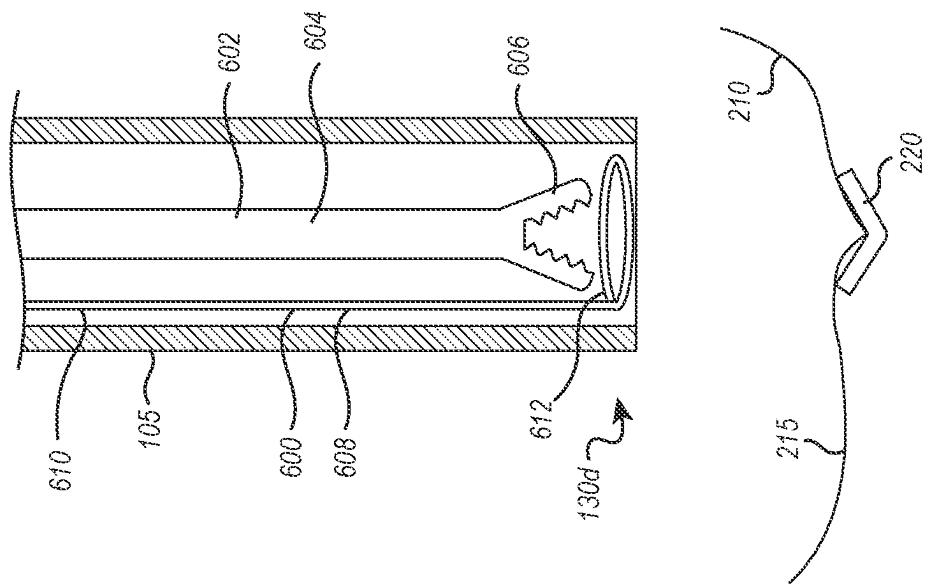

602
604
606
210
220
610
105
600
608
612
130d
215

FIG. 6A

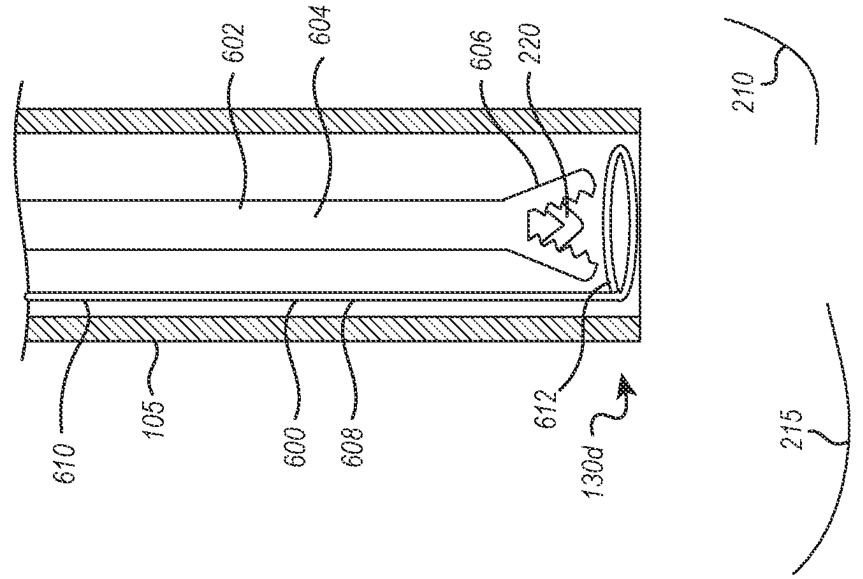

602
604
606
220
210
610
105
600
608
612
130d
215

FIG. 6D

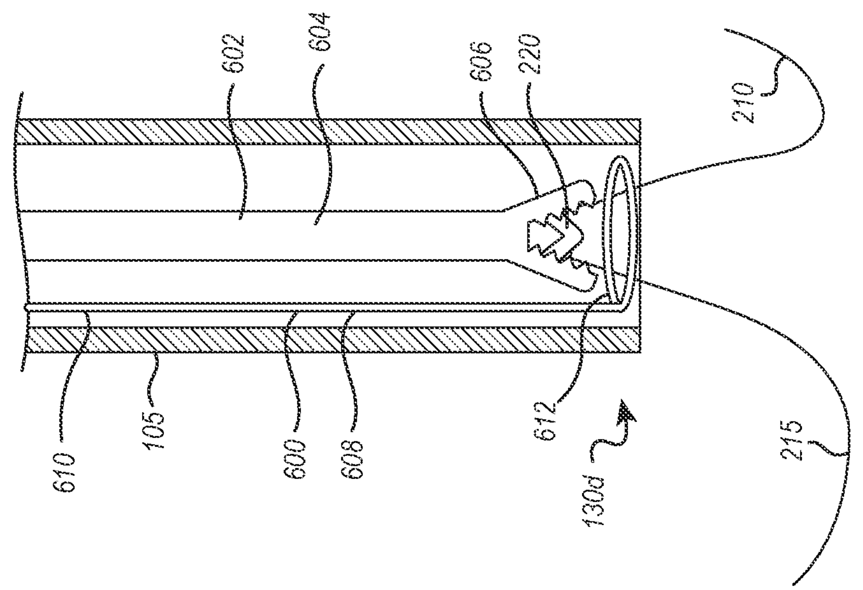

602
604
606
220
210
610
105
600
608
612
130d
215

FIG. 6C

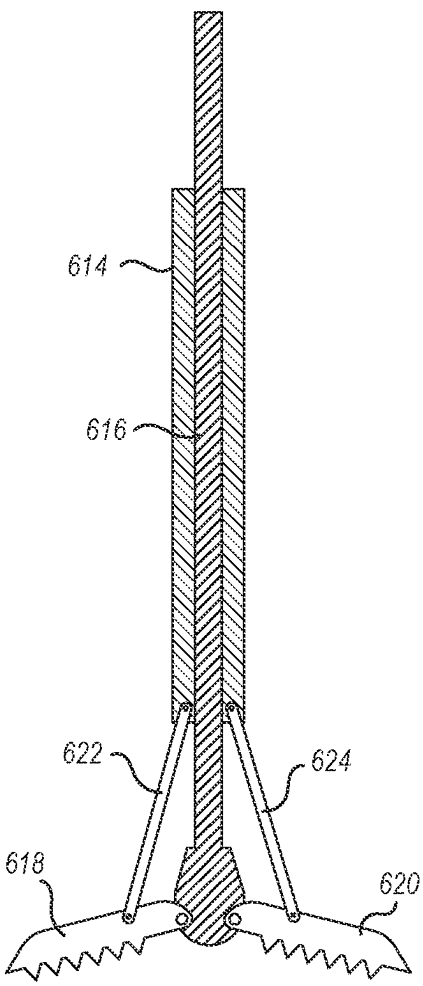


FIG. 6E

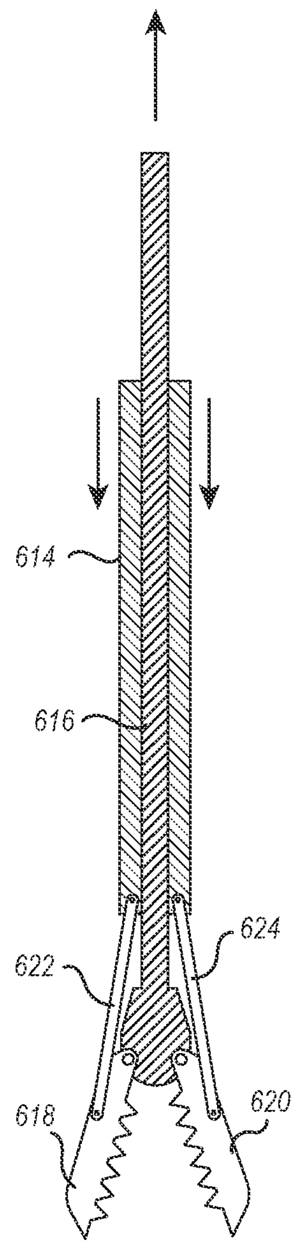


FIG. 6F

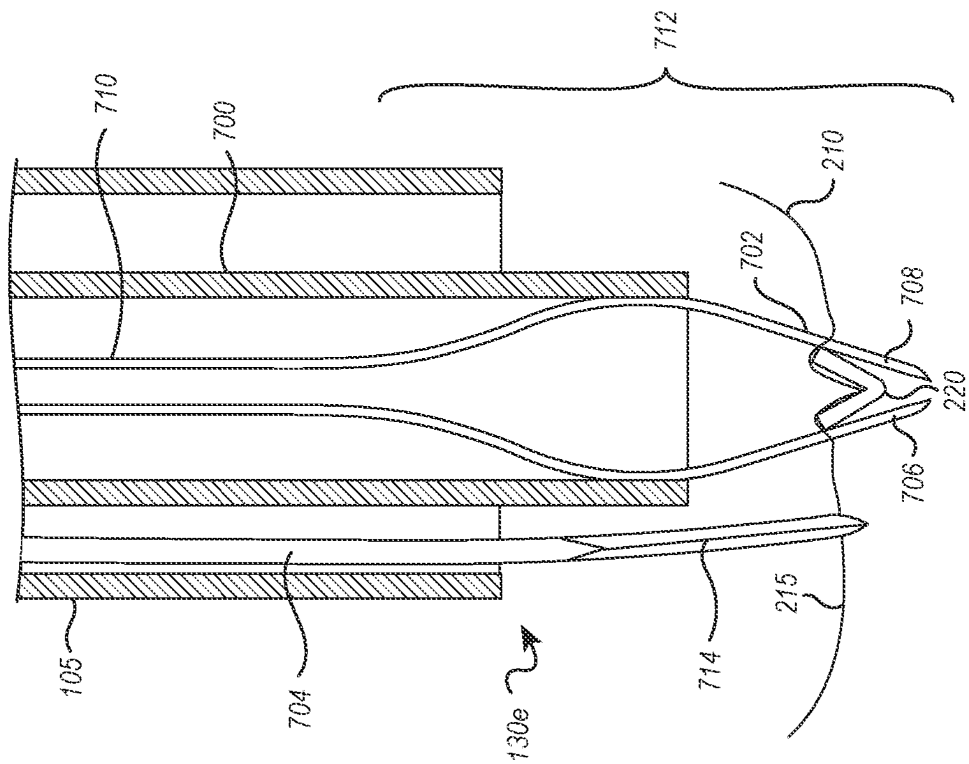

712
710
700
210
702
708
220
706
105
704
130e
714
215

FIG. 7B

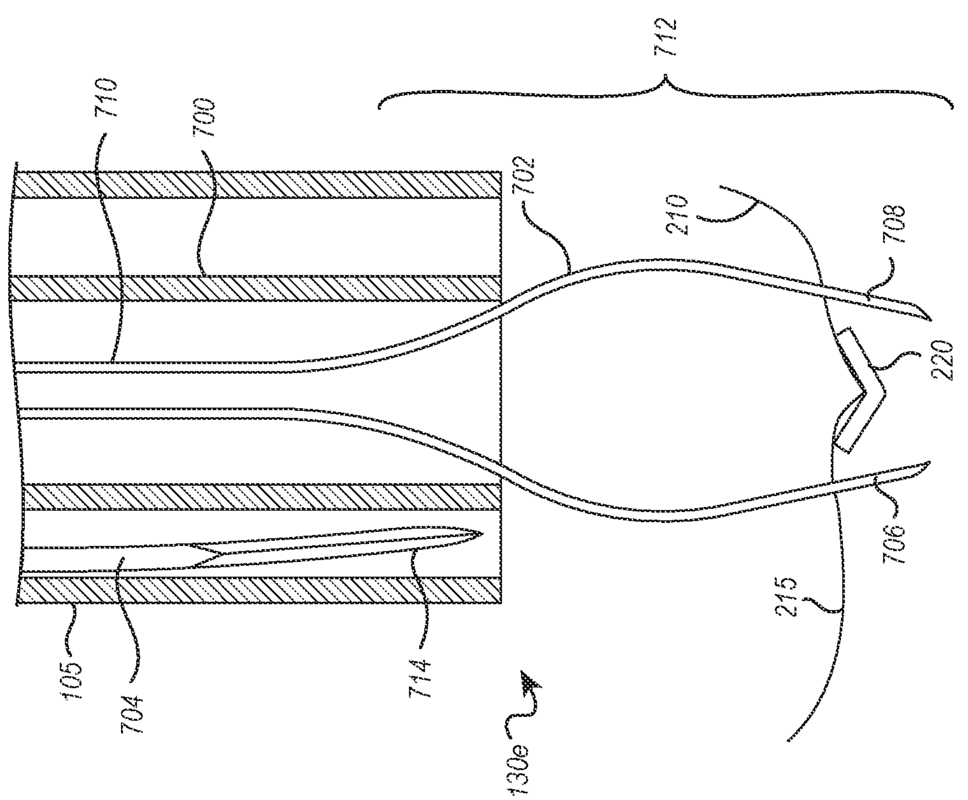

712
710
700
702
210
708
220
706
215
105
704
714
130e

FIG. 7A

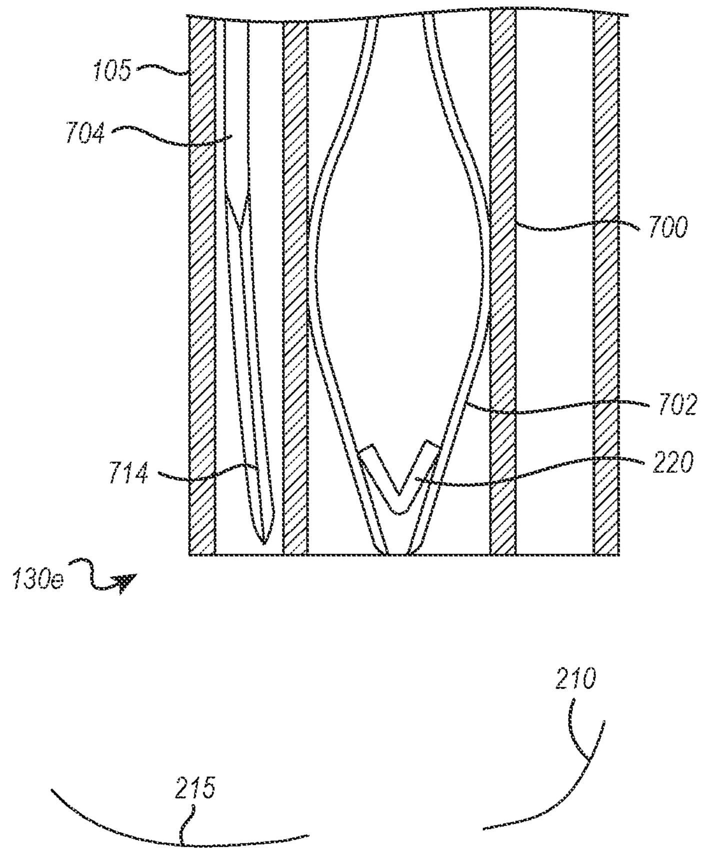


FIG. 7C

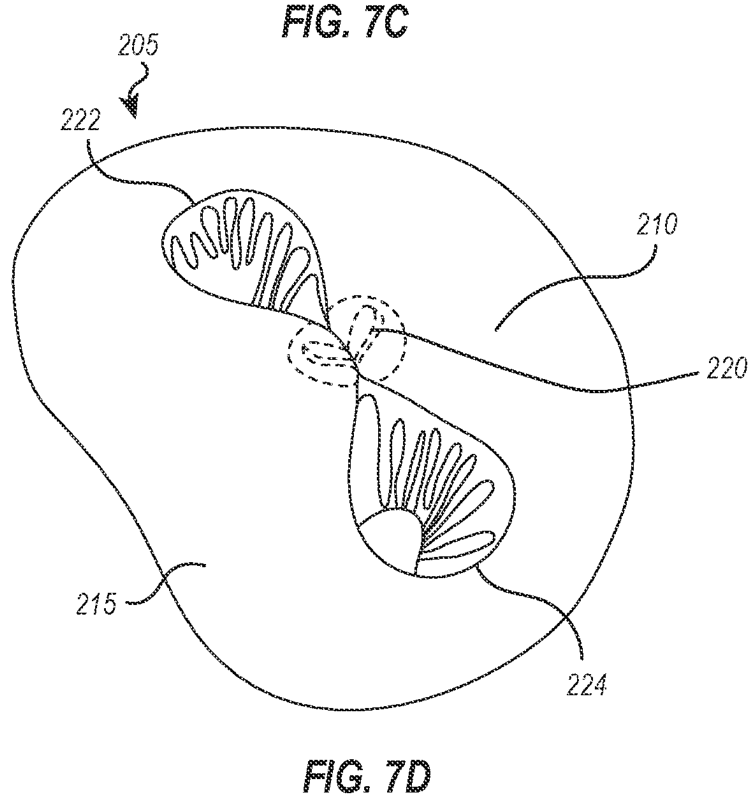


FIG. 7D

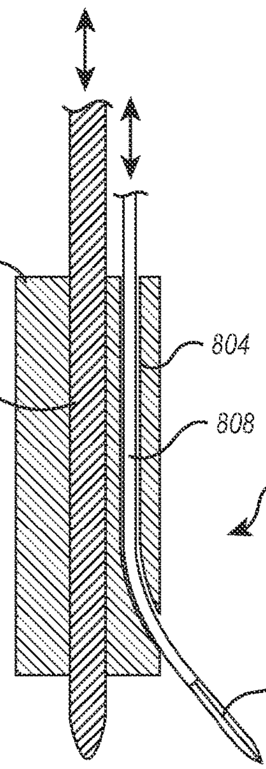


FIG. 8A

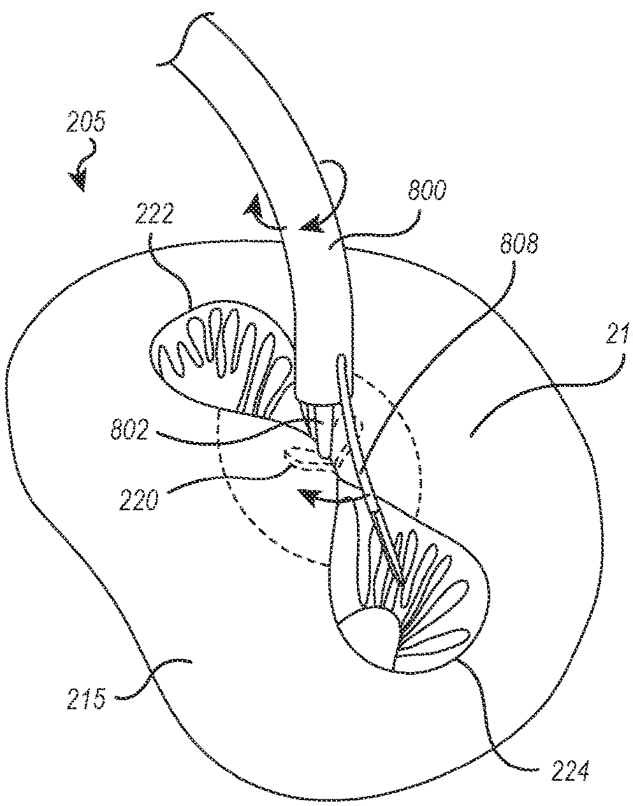


FIG. 8B

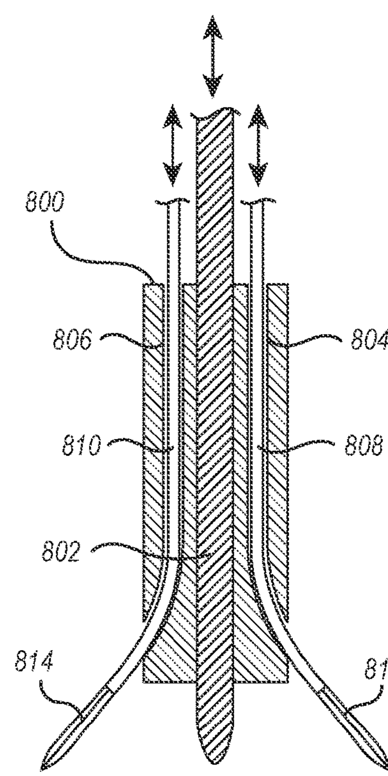


FIG. 8C

DEVICES AND METHODS FOR LEAFLET CUTTING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Patent Application No. 63/020,669, filed May 6, 2020, the entire contents of which are incorporated by reference herein.

BACKGROUND OF THE INVENTION

The mitral valve controls blood flow from the left atrium to the left ventricle of the heart, preventing blood from flowing backwards from the left ventricle into the left atrium so that it is instead forced through the aortic valve for delivery of oxygenated blood throughout the body. A properly functioning mitral valve opens and closes to enable blood flow in one direction. However, in some circumstances the mitral valve is unable to close properly, allowing blood to regurgitate back into the atrium.

Mitral valve regurgitation has several causes. Functional mitral valve regurgitation is characterized by structurally normal mitral valve leaflets that are nevertheless unable to properly coapt with one another to close properly due to other structural deformations of surrounding heart structures. Other causes of mitral valve regurgitation are related to defects of the mitral valve leaflets, mitral valve annulus, or other mitral valve tissues.

The most common treatments for mitral valve regurgitation rely on valve replacement or repair including leaflet and annulus remodeling, the latter generally referred to as valve annuloplasty. One technique for mitral valve repair which relies on suturing adjacent segments of the opposed valve leaflets together is referred to as the "bowtie" or "edge-to-edge" technique. While these techniques can be effective, they usually rely on open heart surgery where the patient's chest is opened, typically via a sternotomy, and the patient is placed on cardiopulmonary bypass. The need to both, open the chest, and place the patient on bypass, is traumatic and has an associated high mortality and morbidity rate. In some patients, a fixation device can be installed into the heart using minimally invasive techniques. The fixation device can hold the adjacent segments of the opposed valve leaflets together and may reduce mitral valve regurgitation. One such device used to clip the anterior and posterior leaflets of the mitral valve together is the MitraClip® fixation device, sold by Abbott Vascular, Santa Clara, Calif., USA.

However, sometimes after a fixation device is installed, undesirable mitral valve regurgitation can still exist, or can arise again. For patients requiring re-intervention, the presence of a fixation device in their mitral valves obstructs transcatheter mitral valve replacement. These patients may also be considered too frail to tolerate open-heart surgery, so they are left with no viable options to further improve the function of their mitral valve.

Accordingly, it would be desirable to provide alternative and additional methods, devices, and systems for removing or disabling fixation devices that are already installed in preparation for the installation of an artificial, replacement mitral valve. The methods, devices, and systems may be useful for repair of tissues in the body other than heart valves. At least some of these objectives will be met by the invention described hereinbelow.

BRIEF SUMMARY OF THE INVENTION

Implementations of the present invention solve one or more problems in the art with systems, methods, and apparatus configured to cut leaflet tissue at a cardiac valve. The system may comprise a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve. The system may also include a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve. Finally, the system may comprise a cutting mechanism, routable through the guide catheter and able to be positioned at the distal end of the guide catheter, configured to cut a portion of leaflet tissue of the cardiac valve.

A method for cutting leaflet tissue at a cardiac valve within a body may comprise positioning a guide catheter, having a proximal and a distal end such that the distal end of the guide catheter is positioned at a cardiac valve. The method may further comprise routing a cutting mechanism through the guide catheter such that the cutting mechanism extends to the distal end of the guide catheter, wherein the cardiac valve is associated with an interventional implant that approximates adjacent leaflets of the cardiac valve, and a cutting mechanism extends from the guide catheter. Also, the method may include positioning the hook catheter to place the cutting mechanism into contact with leaflet tissue located adjacent to the interventional implant and actuating the cutting mechanism to cut at a portion of least one leaflet of the approximated adjacent leaflet.

Additional features and advantages of exemplary implementations of the invention will be set forth in the description which follows, and in part will be obvious from the description, or may be learned by the practice of such exemplary implementations. The features and advantages of such implementations may be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims. These and other features will become more fully apparent from the following description and appended claims or may be learned by the practice of such exemplary implementations as set forth hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to describe the manner in which the above-recited and other advantages and features of the invention can be obtained, a more particular description of the invention briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the invention and are not therefore to be considered to be limiting of its scope, the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

FIG. **1** illustrates an embodiment of a delivery system that may be utilized for guiding and/or delivering a cutting mechanism to a cardiac valve;

FIG. **2** is a schematic representation of a human heart with an interventional implant affixed between the anterior and posterior leaflets of the mitral valve and a guiding catheter extending from a transseptal puncture and positioned above the mitral valve;

FIGS. **3**A-**3**D illustrate an embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve;

FIGS. **4**A-**4**F illustrate an alternative embodiment of a cutting mechanism according to the present disclosure;

FIGS. 5A-5J illustrate an alternative embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve;

FIGS. 6A-6F illustrate an alternative embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve;

FIGS. 7A-7D illustrate alternative embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve; and

FIGS. 8A-8C illustrate yet another embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Implementations of the present invention solve one or more problems in the art with systems, methods, and apparatus configured to cut leaflet tissue at a cardiac valve. More specifically, at least one embodiment of the present invention, may comprise a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve. The system may also include a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve. Finally, the system may comprise a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to cut a portion of leaflet tissue of the cardiac valve.

FIG. **1** illustrates an embodiment of a delivery system **100** that may be utilized for guiding and/or delivering a cutting mechanism to the cardiac valve. In at least one embodiment, the delivery system **100** includes a leaflet cutting system **100**, which can include a delivery system **102** that may be utilized for guiding and/or delivering a cutting mechanism to the cardiac valve. The delivery system **102** can include a guide catheter **105** having a proximal end **112** and a distal end **115**. The delivery system may comprise a handle **110** positioned on the proximal end **112** of the guide catheter **105**. The guide catheter **105** may be operatively coupled to the handle **110**. The guide catheter **105** may include a steerable portion **117** near the distal end **115** that can be steerable to enable the guiding and orienting of the guide catheter **105** through a patient's vasculature to a targeted treatment site, such as a mitral valve. For example, the handle **110** may include at least one control **120** (e.g., a dial, a switch, a slider, a button, etc.) that can be actuated to control the movement and curvature of a steerable portion **117**of the guide catheter **105**.

In at least one embodiment, the at least one control **120** can be operatively coupled to one or more control lines **125** (e.g., pull wires) extending from the handle **110** through the guide catheter **105** to the distal end **115** of the guide catheter (e.g., through one or more lumens in the guide catheter **105**). Actuation of the at least one control **120** may adjust the tensioning of one or more control lines **125** to steer the guide catheter **105** in a desired curvature and/or direction. FIG. **1** shows the handle **110** as having a single control **120** for providing steerability. Alternatively, a handle **110** may comprise more than one control **120** associated with any number of control lines.

While control lines or wires are described at various points in this application, it should be understood that references made throughout this application to control lines or wires may be a single wire or plurality of wires including or made of steel, titanium alloy, aluminum alloy, nickel alloy, other metals, a shape memory material (such as a shape memory alloy or shape memory polymer), inorganic polymer, organic polymer, ceramic, carbon materials, or other flexible material with sufficient tensile strength. For example, a control line **125** may be a steel cable. In another example, a control line **125** may be a monofilament suture. In another example, a control line **125** may be a multifilament suture. In yet another example, a control line **125** may be a braided suture.

It is desirable for guide catheter **105** to provide an adjustable distal end **115**, which is capable of being positioned within a target body cavity in a desired orientation. Guide catheter **105** should have a large lumen diameter to accommodate the passage of a variety of devices, such as the various embodiments of the cutting mechanisms discussed hereinafter, and should have good wall strength to avoid kinking or collapse when bent around tight curves, and should have good column, tensile, and torsional strength to avoid deformation when the devices are passed through the lumen and torqued or translated. Guide catheter **105** should provide for a high degree of controlled deflection at its distal end **115**, but should not take up significant lumen area to allow for passage of interventional devices, such as the cutting mechanisms discussed below. Further, guide catheter **105** should be able to be positioned in a manner which allows compound curves to be formed, for example curvature within more than one plane. Such manipulation should also allow fine control over distal end **115** to accommodate anatomical variations within the same type of body cavity and for use in different types of body cavities.

The guide catheter **105** may comprise a main body made of or including a flexible material. The main body may be made of or include a variety of flexible materials, such as thermoplastic elastomers (TPE). In some embodiments, the main body may be a polyether block amide (PEBA or PEBAX). The main body may have a constant durometer or may have varying durometer that varies along its longitudinal length or that varies in different portions of the body. For example, the main body of guide catheter **105** may be made of or include a body material having a durometer of 25D to 75D. In another example, the main body of guide catheter **105** may be made of or include a body material that has a durometer of about 45D. In at least one embodiment, the body material may include PEBAX **4533**. In at least another embodiment, the body material may include PEBAX **3533**.

The guide catheter **105** preferably defines a central lumen, extending axially through its entire length, through which other elongate elements, such as the cutting mechanisms may be inserted for accessing a treatment site. The central lumen may also include a central lumen lining on an inner surface thereof. In some embodiments, the central lumen lining may be a protective material that protects the interior walls from damage due to another element of the elongated member moving through or within the central lumen. In other embodiments, the central lumen lining may include a lubricious coating that reduces friction between the interior wall and another element of the elongated member moving through or within the central lumen. The central lumen lining may include PEB A, polytetrafluoroethylene ("PTFE"), polyetheretherketone ("PEEK"), other polymers, thermoplastic polyurethane ("TPU"), polyethylene with pebble stone surface, silicone oil stainless steel, Nitinol, other metals, or combinations thereof. In at least one embodiment, the central lumen lining may include a plurality of PEBA materials having different durometers.

In other embodiments, the guide catheter **105** may also have an outer layer. In some embodiments, the outer layer may be made of or include a single material or may be made of or include different materials to impart different handling characteristics to the guide catheter **105**. For example, the outer layer may be made of or include softer materials to promote flexibility of the guide catheter **105**. In other examples, the outer layer may be made of or include stiffer materials to promote pushability and/or torqueability of the guide catheter **105**. In yet other examples, the outer layer may include lubricious materials to reduce friction between the guide catheter **105** and the body lumen of the patient. The outer layer may include PEBA, polytetrafluoroethylene ("PTFE"), polyetheretherketone ("PEEK"), other polymers, thermoplastic polyurethane ("TPU"), polyethylene with pebble stone surface, silicone oil stainless steel, Nitinol, other metals, or combinations thereof. In at least one embodiment, the outer layer may include a plurality of PEB A materials having different durometers.

In some embodiments, the outer layer of guide catheter **105** may also include a radiopaque marker to improve visualization of guide catheter **105** during a medical procedure. For example, the outer layer may include a barium sulfate (BaSO4), gold, platinum, platinum iridium, iodine, other radiopaque materials, or combinations thereof on a distal portion of guide catheter **105**. In at least one embodiment, one or more additional radiopaque markers may be longitudinally located at one or more intermediate locations along the length of guide catheter **105**.

The curves of guide catheter **105** may be formed by any suitable means. In some embodiments, one or more of the curves are preset so that the curve is formed by shape memory. For example, guide catheter **105** may be comprised of a flexible polymer material in which a curve is preset by heating. When guide catheter **105** is loaded on a guidewire, dilator, obturator or introductory device, the flexibility of guide catheter **105** can allow it to follow the shape or path of the introductory device for proper positioning within the body. When the introductory device is pulled back and/or removed, guide catheter **105** can then resume the shape memory configuration which was preset into the catheter.

Alternatively, the curves may be formed or enhanced with the use of one or more steering mechanisms. In some embodiments, the steering mechanism comprises at least one control wire or pull wire attached to one of the guide catheter **105**, wherein actuation of the steering mechanism applies tension to the at least one pull wire whereby the curve is formed. The pull wires can extend through the central lumen or through individual lumens in the wall of guide catheter **105**. It may be appreciated that more than one pull wire may extend through any given lumen. The presence of each pull wire allows curvature of guide catheter **105** in the direction of the pull wire. For example, when pulling or applying tension to a pull wire extending along one side of the catheter, the catheter will bend, arc or form a curvature toward that side. To then straighten the catheter, the tension may be relieved for recoiling effects or tension may be applied to a pull wire extending along the opposite side of the catheter. Therefore, pull wires are often symmetrically placed along the sides of the catheter.

Thus, in some embodiments at least two pull wires are attached in diametrically opposed locations wherein applying tension to one of the pull wires curves the catheter in one direction and applying tension to the pull wire attached in the diametrically opposed location curves the catheter in another direction opposite to the one direction. The diametrically opposed pull wires may be considered a set. Any number of sets may be present in a catheter to provide unlimited directions of curvature. In some embodiments, the steering mechanism can comprise at least four pull wires wherein two of the at least four pull wires are attached to the guide catheter in diametrically opposed locations and another two of the at least four pull wires are attached to the guide catheter in diametrically opposed locations. In other words, the catheter may include two sets of pull wires, each set functioning in an opposing manner as described. When the two sets of pull wires are positioned so that each pull wire is 90 degrees apart, the catheter may be curved so that the distal end is directed from side to side and up and down. In other embodiments, the steering mechanism comprises at least three pull wires, each pull wire symmetrically positioned approximately 120 degrees apart. When tension is applied to any of the pull wires individually, the catheter is curved in the direction of the pull wire under tension. When tension is applied to two pull wires simultaneously, the catheter is curved in a direction between the pull wires under tension. Additional directions may also be achieved by various levels of tension on the pull wires. It may be appreciated that any number, combination and arrangement of pull wires may be used to direct the catheters in any desired direction.

In some embodiments, a portion of one of guide catheter **105** can comprise one or more articulating members. In this case, the at least one pull wire is attached to one of the articulating members so that the curve is formed by at least some of the articulating members. Each pull wire is attached to the catheter at a location chosen to result in a particular desired curvature of the catheter when tension is applied to the pull wire. For example, if a pull wire is attached to the most distal articulating member in the series, applying tension to the pull wire will compress the articulating members proximal to the attachment point along the path of the pull wire. This results in a curvature forming in the direction of the pull wire proximal to the attachment point. It may be appreciated that the pull wires may be attached to any location along the catheter and is not limited to attachment to articulating members. Typically, the articulating members comprise inter-fitting domed rings but may have any suitable shape.

It may also be appreciated that curves in guide catheter **105** may be formed by any combination of mechanisms. For example, a portion of guide catheter could form a curve by shape memory while a different portion of guide catheter could form a curve by actuation of a steering mechanism.

The steering mechanisms may be actuated by manipulation of actuators located on handle **110**. The handle **110** can be connected with the proximal end of the guide catheter **105** and remains outside of the body. One or more actuators or controls **120** can be provided on handle **110** and may have any suitable form, including buttons, levers, knobs, switches, toggles, dials, or thumbwheels, to name a few. When pull wires are used, each actuator may apply tension to an individual pull wire or to a set of pull wires. The handle may also include one or more locking mechanisms configured to interface with, and selectively lock into place, one or more of the controls **120**.

In at least one embodiment, the handle **110** includes at least one control **120** for actuating and/or adjusting one or more components of a cutting mechanism **130**. As shown in FIG. **1**, the cutting mechanism **130** is configured to extend beyond the distal end **115** of the guide catheter **105**. In at least one embodiment, the cutting mechanism **130** is routable through the guide catheter **105** and retractable into the guide catheter **105**. The at least one control **120** may control the cutting mechanism's **130** extension from, and retraction into, the guide catheter **105**. Additionally or alternatively, the at least one control **120** may be configured to provide selective actuation of the cutting mechanism **130**. The at least one control **120** may be operatively connected to one or more additional elements of the cutting mechanism **130**. The cutting mechanism **130** is shown here in generic form as a dashed line, and therefore represents any of the cutting mechanism **130** embodiments described herein.

FIG. **2** is a schematic representation of a human heart with an interventional implant affixed between the anterior and posterior leaflets of the mitral valve. The mitral valve comprises a posterior mitral leaflet **210** and an anterior mitral leaflet **215**. FIG. **2** also shows an interventional implant (e.g., MitraClip®) **220**, which has previously been affixed to the leaflets in an effort to reduce regurgitation by approximating the adjacent leaflets **210** and **215**. As shown, the affixation of implant **220** to the leaflets creates a first orifice **222** and a second orifice **224** located on opposing sides of implant **220** and between the anterior mitral leaflet **215** and the posterior mitral leaflet **210**. And, as discussed above, if further treatment is required in the form of the installation of an artificial, replacement mitral valve, the prior clip implant **220** must be detached from one or both of the leaflets before the replacement valve can be implanted.

In use, the leaflet cutting system **100** can be inserted into, and navigated through a patient's vasculature in a conventional manner to arrive in the patient's heart. As also illustrated in FIG. **2**, a distal end portion of the leaflet cutting system **100** can be inserted through the interatrial septum **200** of the heart and positioned above the mitral valve in preparation for a leaflet cutting procedure. In at least one embodiment, the distal end **115** of the guide catheter **105** can include one or more radiopaque or echogenic markers to aid in accurate positioning. One skilled in the art will appreciate that the positioning of the distal end portion of the guide catheter **105** of the leaflet cutting system **100** in FIG. **2** is merely exemplary, and the present disclosure is not limited to the specific positioning shown.

Referring next to FIGS. **3A-3D**, which illustrate a first embodiment of a leaflet cutting system **100**. System **100** includes components corresponding to those described above in connection with FIG. **1**, which are designated by the same reference numerals throughout. As shown in FIGS. **3A-3D**, system **100** also includes a cutting mechanism **130*a*** positioned within, and routed through, the guide catheter **105**. As shown, the cutting mechanism **130*a*** may comprise an elongate inner catheter or hypotube **300**, extending from a proximal end (not shown in FIG. **3**) to a distal end **302** of the leaflet cutting system **100**. The proximal end of the cutting mechanism **130*a*** can be operatively coupled to the handle **110**, and handle **110** provided with controls adapted to manipulate the cutting catheter **130*a***, including advancing, retracting and/or rotating the cutting mechanism **130*a*** relative to the guide catheter **105**.

As shown in FIG. **3A**, cutting mechanism **130*a*** is positioned within the interior of guide catheter **105** during advancement and positioning of the system **100** above the mitral valve **205**. As discussed below, once the distal end **115** of guide catheter **105** is properly positioned above the mitral valve **205** and in proper alignment with interventional implant **220**, the handle **110** can be manipulated to cause cutting mechanism **130*a*** to advance relative to guide catheter **105** and to pass through the tissue of one of the leaflets **210**, **215**, thereby cutting the affected leaflet and separating the interventional implant **220** from the affected leaflet.

In this embodiment, the cutting mechanism **130*a*** can have at its distal end **302**, a sharped end **304** that terminates in a point. The sharpened end **304** may have a circular or oval cross-sectional shape. The sharpened end **304** can also have tapered, sharpened edges or blades adjacent to and extending from the point, which are adapted to slide through the leaflet tissue located between the first and second orifices **222**, **224** and adjacent the interventional implant **220**. In this and other embodiments described herein, the components of cutting mechanism **130*a*** may be formed from the same or different materials, including but not limited to stainless steel or other metals, Elgiloy®, nitinol, titanium, tantalum, metal alloys or polymers.

FIG. **3B** also shows how the hypotube **300** may be advanced to cause the sharpened end **304** to extend from the distal end **115** of the guide catheter **105** and cut a portion of the anterior mitral leaflet **215**. The point of sharpened end **304** helps anchor and stabilize the leaflet tissue as the sharpened, cutting edges of the cutting mechanism **130*a*** are advanced through and cut the leaflet tissue. The circular or oval cross section of sharpened end **304** of the cutting mechanism **130*a*** cause the sharpened end to cut through the leaflet tissue in an arc around the interventional implant and extending from the first orifice **222** to the second orifice **224**, as schematically illustrated in FIG. **3D**. After the sharpened end **304** of cutting mechanism **130*a*** advances and cuts through the tissue of the anterior mitral leaflet **215**, the interventional implant **220** may remain attached to the posterior mitral leaflet **210**, as shown in FIG. **3C**, thereby reducing the risk that the interventional implant **220** will interfere with functioning of the left ventricular outflow tract. Additionally or alternatively, the posterior mitral leaflet **210** may also be cut in a similar manner with little or acceptable risk of left ventricular outflow tract interference. In at least one embodiment, the interventional implant **220** is removed from the patient.

FIG. **3D** is a top perspective view of a mitral valve showing the cut portion of the anterior mitral leaflet **215**. As shown, the sharpened end **304** may cut an arc in the leaflet **210** around the interventional implant **220**. The hypotube **300** may be sized depending on the number of interventional implants **220** that are located within the cardiac valve. Additionally or alternatively, a single hypotube **300** may be used to cut multiple interventional implants **220** within a cardiac valve.

In this and other embodiments described herein, inner catheter or hypotube **300** should preferably have sufficient flexibility as to be able to conform to bends formed by guide catheter **105**. Additional flexibility to accommodate bending may be provided in certain regions of hypotube **300** by a series of laser cuts formed in the outer wall of hypotube **300**. In addition, hypotube **300** should also provide sufficient compressive strength to permit forces to be transmitted through hypotube **300**, from the proximal end to the distal end, sufficient to cause the sharpened end **304** of hypotube **300** to cut through the leaflet tissue.

In an alternate embodiment, an indeflator **240** can be attached to the guide catheter **105** at or near its proximal end. The indeflator **240** can be configured to create a vacuum transmitted through the lumen of the guide catheter **105**. When the distal end of the guide catheter is positioned against the leaflet, a negative pressure can be applied to the leaflet tissue, thereby stabilizing nearby leaflet tissue located adjacent the interventional implant **220**. While leaflet tissue adjacent to the clip implant **220** is held in place by the negative vacuum pressure applied through the distal end of the guide catheter, the hypotube **300** can be advanced relative to the guide catheter **105**, thereby causing the sharpened end **304** to pass through and cut the leaflet tissue surrounding the clip implant **220**. In this alternate embodiment, a distal end portion of the guide catheter **105** may comprise various cross-sectional shapes including a U-shape. For example, the distal end **115** of the guide catheter **105** can comprise a flexible U-shape that may collapse when positioned within a delivery sheath and then expand when the guide catheter is advanced to extend beyond the distal end of the delivery sheath.

In another alternate embodiment, in addition to the sharpened edges located at the distal end of the cutting mechanism **130***a* used for mechanical cutting, the cutting mechanism **130***a* could also include an electrical conductor (not shown) that extends along its entire length, which conductor is electrically coupled at a distal end to the sharpened cutting end **304** and which is also electrically coupled at a proximal end to a source for selectively applying electrosurgical energy, such as an electrosurgical generator. In that case, the cutting of leaflet tissue could be achieved by mechanical cutting, by the application of electrosurgical energy, or a combination of both.

Reference is next made to FIGS. **4**A-**4**F, which illustrate a second embodiment of a cutting mechanism **130***b*. This embodiment also includes components corresponding to those described above in connection with FIG. **1**, which are designated by the same reference numerals throughout. FIG. **4**A is a side sectional view of the cutting mechanism **130***b* shown within the distal end **115** of the guide catheter **105**. As shown, the cutting mechanism **130***b* may comprise an inner catheter or hypotube **400** having a sharpened, cutting end **402**, which terminates at its distal end in a sharpened point. This embodiment, the sharpened cutting end **402** can include two blades **404** and **406** joined at a pivot point. The two blades **404** and **406** may each comprise a cutting edge disposed on an outer edge of each blade as illustrated in FIG. **4**B. The hypotube **400** may be configured to be selectively extended from the distal end **115** of the guide catheter **105**, thereby allowing the two blades **404** and **406** to engage with a portion of leaflet tissue.

In this embodiment, the blades **404** and **406** can be connected to the hypotube **400** in such a way that, when actuated, the blades pivot outwardly away from one another in an arcuate path in a "reverse-scissor" fashion. In the illustrated embodiment, this can be accomplished as follows. A tip ring **408** can be integrally attached to the distal end of the hypotube. Tip ring **408** can include a center lumen that allows an actuating rod **410** to extend therethrough as shown in FIG. **4**B. The actuating rod **410** can be pivotally connected at its distal end to a center portion of the proximal ends of each of blades **404** and **406** as shown. Links **412** and **414** can also be provided, which connect a distal end of tip ring **408** to an outer portion of the proximal ends of each of blades **404** and **406**. While not explicitly shown in FIG. **4**, hypotube **400** and actuating rod **410** can run the entire length of delivery system **102** and can be operatively coupled at their proximal ends to handle **110**, and handle **110** can be provided with suitable controls to allow manipulation of hypotube **400** and actuating rod **410** relative to one another to cause blades **404** and **406** to open and close relative to one another. It will be appreciated that advancing actuating rod **410** relative to hypotube **400** will cause blades **404** and **406** to open in an arcuate path away from one another, and that withdrawing actuating rod **410** relative to hypotube **400** will cause blades **404** and **406** to close and move toward one another. Alternatively, actuating rod **410** could be held stationary and hypotube **400** moved relative to actuating rod **410** to open and close blades **404** and **406**. Other structures and mechanisms known to those skilled in the art could be adapted and used for opening and closing blades **404** and **406** and are intended to be encompassed within the scope of the invention.

The cross-sectional shape of blades **404** and **406** can be selected to influence that shape of the path they travel as they cut through leaflet tissue. For example, in one embodiment illustrated in FIG. **4**C, blades **404** and **406** can have a relatively straight or flat cross-sectional shape, which will cause the blades to cut through the leaflet tissue in a relatively straight path. Referring to FIG. **4**D, the cut through the leaflet tissue can be made in multiple segments. For example, with blades **404** and **406** closed, the sharpened end **402** can be inserted at a first insertion point **416***a* into and through leaflet **210**. The sharpened point and sharpened edges of sharpened end **402** assist with the ease of insertion and also help stabilize the tissue adjacent to the interventional implant **220**. Once inserted through the tissue, blades **404** and **406** can be actuated by means of handle **110**, acting through hypotube **400** and actuating rod **410**, to cause the blades to open, thereby cutting through the leaflet tissue located to either side of insertion point **416***a*. The cutting mechanism can then be withdrawn in a proximal direction, repositioned, re-advanced in a distal direction through the tissue at a second insertion point **418***a*, re-actuated, and then at a third insertion point **418**c, and so and so forth until the process completed to create a continuous cut through the affected leaflet tissue around the interventional implant **220** and extending from the first orifice **222** to the second orifice **224**. Alternatively, the opening and closing of blades **404** and **406** can simply be reciprocated as the guide catheter **105** is used to drag the cutting blades through the leaflet tissue.

In an alternate embodiment, as illustrated in FIGS. **4**E-**4**F, the two blades **404** and **406** can be shaped such that they cut the portion of leaflet tissue in an arcuate path. As best shown in FIG. **4**E, blades **404** and **406** can also be made to have a cross-sectional shape that is circular or oval in shape. As with the above, the pointed end of sharpened end **402** is advanced through an entry point **416***a* into the leaflet tissue. However, when actuated, the shape of blades **404** and **406** cause them to travel in an actuate path through the leaflet tissue, as illustrated in FIG. **4**F.

The cut portion in FIG. **4**F is shown on the anterior leaflet **215**, however, the present invention is not limited to the positioning of shape or cut. In at least one embodiment, the posterior mitral leaflet **210** can also be cut. Additionally or alternatively, the interventional implant **220** may be removed from the patient.

Reference is next made to FIGS. **5**A-**5**J, which illustrate yet another embodiment of the cutting mechanism. In this embodiment, cutting mechanism **130***c* may comprise a first and a second delivery catheter **500** and **502**, each comprising a proximal end and a distal end. The first delivery catheter **500** can include at its distal end a pair of grasping arms **504** and **506**, which arms can be rotatably coupled to the distal end of delivery catheter **500** by a rotatable hub **508**. Similarly, second delivery catheter **502** can include at its distal end a pair of grasping arms **510** and **512**, which arms can be rotatably coupled to the distal end of delivery catheter **502** by a rotatable hub **514**. One or more of grasping arms **504**, **506**, **510** and/or **512** may include one or more teeth to prevent slippage when the arms are engaged with leaflet tissue.

A cutting wire **516** may extend from the hub **508** of the first delivery catheter **500** to the **514** of the second delivery catheter **502**. The cutting wire **516** may be configured to selectively provide electrosurgical energy to a secured portion of leaflet tissue, thereby cutting a portion of leaflet tissue when rotated around hubs **508** and **514**. Additionally or alternatively, an adjustment in the tension of cutting wire **516** may cause the wire **516** to engage with and cut the secured leaflet tissue.

As shown in FIG. **5**B, which is a top perspective view of a mitral valve, the first and second delivery catheters **500** and **502** may be configured to extend from the distal end **115** of the guide catheter **105**. The interventional implant **220** creates a first orifice **222** and a second orifice **224** between the anterior mitral leaflet **215** and the posterior mitral leaflet **210** by approximating the adjacent leaflets **210** and **215**. The first delivery catheter **500** may extend toward the first orifice **222** and thereby extend the first grasping arm **504** (not shown in FIG. **4**B) of the first delivery catheter **500** into the first orifice **222** of the mitral valve. Similarly, the second delivery catheter **502** may extend toward the second orifice **224** and thereby extend the first grasping arm **510** (not shown in FIG. **4**B) of the second delivery catheter **502** into the second orifice **224** of the mitral valve. The first grasping arms **504** and **510** may be configured to engage with leaflet tissue on the ventricular side of the mitral valve.

FIGS. **5**C and **5**D are a detailed side views of the distal end of delivery catheter pair **500** and **502** at various stages of positioning and deployment to attach and hold onto a portion of the tissue of one of the leaflets **205**, with one delivery catheter **500** attaching to the leaflet **210** to one side of the interventional implant **220** and the other delivery catheter **502** attaching to the same leaflet **210** on the other side of the interventional implant **220**. Once the guide catheter **105** is properly positioned above and aligned with the interventional implant **220**, delivery catheters **500** and **502** are advanced relative to the guide catheter **105** to extend beyond the distal end **115** of guide catheter **105**. The distal ends of delivery catheters **500** and **502** are advanced until grasping arms **504** and **510** pass through orifices **222** and **224** of the mitral valve on opposites side of the interventional implant **220**, as shown in FIG. **5**C. Guide catheter **105** can then be manipulated to reposition grasping arms **504** and **510** into direct contact with the edges of orifices **222** and **224**, as also shown in FIG. **5**C. Once that is done, grasping arms **504** and **506** of delivery catheter **500** can be rotated toward one another to clamp down on the leaflet tissue adjacent the first orifice **222** (and located to one side of the interventional device **220**), and grasping arms **510** and **512** of delivery catheter **502** can similarly be rotated toward one another to clamp down on either side of the leaflet tissue adjacent the second orifice **224** (and located to the other side of the interventional device **220**), as shown in FIG. **5**D. Once the leaflet tissue is secured between grasping arms **504** and **506** and between grasping arms **510** and **512**, then cutting wire **516** can be rotated to cut through the leaflet tissue and separate the interventional implant **220** from the affected leaflet, as illustrated in FIGS. **5**E and **5**F.

FIGS. **5**G-**5**J illustrate one example of a mechanism of actuating the various components of cutting mechanism **130***c* described above. However, additional structures and mechanisms known to those skilled in the art could be adapted and used for controlling the selective, relative rotation of grasping arms **504**, **506**, **510** and **512**, as well as that of cutting wire **516**. In one embodiment, each of delivery catheters **500** and **502** can include at their respective distal ends a tip ring **518**, having a plurality of lumens for accepting a plurality of control lines. Each control line, such as control line **522** shown in FIGS. **5**H and **51**, can be coupled at the proximal end of the system **100** to a dedicated control located on handle **110**. Separate control lines can be provided to separately and independently control selective movement of each of grasping arms **504**, **506**, **510** and **512**, as well as movement of cutting wire **516**. Certain control lines, such as control line **522**, can extend the entire length of the system and pass around a saddle, such as saddle **520** formed in tip ring **518** between adjacent lumens, as shown in FIGS. **5**I and **5**J, so that the control line returns in a proximal direction for attachment to one of the rotatable hubs, such as rotatable hub **508**, as shown in FIG. **5**J. As further shown in FIG. **5**J, control line **522** can be mechanically coupled to the periphery on one side of hub **508**, and a second control line **524** can be mechanically coupled to the periphery of the other side of hub **508**. As will be readily understood, selectively pulling on control line **522** will cause hub **508** to rotate in clockwise direction, and selectively pulling on control line **524** will cause hub **508** to rotate in counter-clockwise direction. By providing multi-part hubs and separate control lines for each component, relative movement and control of grasping arms **500** and **502**, grasping arms **510** and **512**, and cutting wire **516** can be effectuated from the proximal end of system **100** by means of one or more controls **120** provided on handle **110**.

FIGS. **5**C-**5**F illustrate how the grasping arms **504**, **506**, **510** and **512** are configured to secure a portion of the anterior mitral leaflet **215** therebetween and show the cutting wire **516** as cutting a portion of the anterior mitral leaflet **215**. However, in at least one embodiment the posterior mitral leaflet **210** can be cut. The interventional implant **220** may additionally or alternatively be removed. As shown in FIGS. **5**E and **5**F, the cutting wire **516** may be rotated through the secured portion of the anterior mitral leaflet **215**, thereby cutting the portion of leaflet tissue.

In at least one embodiment, the space between the first and second delivery catheters **500** and **502** may be adjusted based on number of interventional implants **220** within the mitral valve. Additionally or alternatively, the cutting mechanism **130***c* may be configured to retract into the distal end **115** of the guide catheter **105** after the portion of leaflet tissue is cut. One skilled in the art will appreciate that the positioning shown in FIGS. **5**C-**5**F is merely exemplary and the present invention is not limited to the positioning shown.

FIGS. **6**A-**6**F are side sectional views and illustrate yet another alternative exemplary embodiment of a cutting mechanism shown in use in association with a cardiac valve, specifically a mitral valve. In this embodiment, the cutting mechanism **130***d* can include a clip grasping structure **602** and a cutting wire **608**. FIG. **6**B shows the distal end **115** of the guide catheter **105** positioned above the interventional implant **220**, which approximates the anterior mitral leaflet **215** and posterior mitral leaflet **210**. The cutting wire **608** can include an elongated portion **610** that terminates in a distal lasso or loop portion **612**. The cutting wire **608** may extend from the handle **110** to the distal end **115** of the guide catheter **105**. Further, the cutting wire **608** may be configured to selectively provide electrosurgical energy through the distal loop portion **612** to tissue in contact with distal loop portion **612**. While the cutting wire **608** is shown as a separate component in FIGS. **6**A-**6**D, a cutting electrode could also be integrally formed on the distal end **115** of guide catheter **105** and/or on the interior wall surface of guide catheter **105** adjacent the distal end **115**.

FIGS. **6**A-**6**F also illustrate a clip grasping structure **602** comprising an elongated portion **604** and a distal clamping portion **606**, which may be routed through the guide catheter **105**. As shown in FIG. **6**B, the distal clamping portion **606** is extendable distally past the distal end **115** of the guide catheter **105**. The distal clamping portion **606** may extend through the distal loop portion **612** of the cutting wire **608** when extending distally past the distal end **115** of the guide catheter **105**.

The distal clamping portion **606** may be configured to secure an interventional implant **220**, as shown in FIG. **6**C. The distal clamping portion **606** may also be retractable proximally into the distal end **115** of the guide catheter **105** with the secured interventional implant **220**. In at least one embodiment, the distal clamping portion **606** extends through the distal loop portion **612** of the cutting wire **608** when retracting proximally into the distal end **115** of the guide catheter **105** with the secured interventional implant **220**. As shown in FIG. **6**D, the distal loop portion **612** of the cutting wire **608** may be configured to cut a portion of leaflet tissue when the distal clamping portion **606** of the clip grasping structure **602** retracts proximally into the distal end **115** of the guide catheter **105** with the secured interventional implant **220**. The distal loop portion **612** of the cutting wire **608** may also be selectively constricted to engage with and cut the portion of leaflet tissue. In at least one embodiment, the cutting wire **608** is configured to detach the interventional implant **220** from surrounding leaflet tissue.

FIGS. **6**E and **6**F illustrate one example of a mechanism of actuating the distal clamping portion **606** of grasping structure **602**. However, additional structures and mechanisms known to those skilled in the art could be adapted and used for selectively opening, closing and clamping distal clamping portion **606**. In this embodiment, grasping mechanism **602** can include an elongate hypotube **614**, an actuating rod **606**, a pair of grasping arms **618** and **620**, and a pair of links **622** and **624**. A proximal end of each grasping arm **618** and **620** can be pivotally coupled to the distal end of actuating rod **616**. Links **622** and **624** can each be pivotally connected at one end to the distal end of hypotube **614** and can be pivotally connected at the other end to an intermediate location on grasping arms **618** and **620**, respectively. It should be further understood that hypotube **614** and actuating rod **616** can extend the entire length of the system and can be operatively coupled at their proximal ends to suitable controls **120** located on handle **110** to control movement of hypotube **614** and actuating rod **616** relative to one another and, thus, causing grasping arms to open and close. The tissue engaging surface of grasping arms **618** and **620** can also include one or more teeth to engage leaflet tissue adjacent the interventional implant and prevent slippage.

Reference is next made to FIGS. **7**A-**7**D, which illustrate still yet another embodiment of a cutting mechanism for use in detaching an interventional device from one or more leaflets of a mitral valve. Cutting mechanism **130***e* can include an inner catheter or hypotube **700**, a clip grasping structure **702** and an elongated cutter **704**. As shown in Figures**7**A-**7**C, the distal end **115** of the guide catheter **105** may be positioned above the interventional implant **220**. The elongated cutter **704** can be configured to be positioned within the lumen formed between the outer wall of the hypotube **700** and the inner wall of the guide catheter **105** and to selectively extend from the distal end **115** of the guide catheter **105**. Cutter **704** can terminate at is distal end in a sharpened point adapted to pierce through leaflet tissue. Cutter **704** can also include one or more sharpened edges **714** (adapted to mechanically cut through the leaflet tissue) that extend along one or both sides of a distal section thereof. The elongated cutter **704** may rotate around the clip grasping structure in an arcuate path, thereby slicing through a portion of leaflet tissue. In at least one embodiment the elongated cutter **704** can be composed of a heat shaped material.

FIGS. **7**A-**7**C further illustrate that hypotube **700** extends through the guide catheter **105** and can be positioned at the distal end **115** of the guide catheter **105**. Clip grasping structure **702** may be routable through the hypotube **700** and positioned at a distal end of the hypotube **700**. In at least one embodiment, the clip grasping structure **702** can comprise two flexible prongs **706** and **708**, each of which preferably includes a sharpened tip adapted to pierce through the tissue of the mitral valve leaflets. The clip grasping structure **702** may comprise an elongated portion **710** and a distal clamping portion **712**. As shown in FIGS. **7**A and **7**B, the distal clamping portion **712** of the clip grasping structure **702** may be extendable distally past the distal end **115** of the guide catheter **105** and the distal end of the hypotube **700**. In at least one embodiment, the distal clamping portion **712** can be made of a semi-rigid, resilient metal. Once advanced beyond the distal end and freed from the constraints of hypotube **700**, the distal clamping portion **712** can expand laterally as illustrated in FIG. **7**A. Once in this configuration, the distal clamping portion **712** of the clip grasping structure **702** may be positioned around the interventional implant **220**, and the assembly may then be advanced distally, causing the prongs **706** and **708** of distal clamping portion **712** to puncture through leaflet tissue located to either side of the interventional device **220**. Then, as shown in FIG. **7**B, hypotube can be advanced distally from the distal end **115** of the guide catheter **105** relative to the clip grasping structure **702**, such that the hypotube **700** encloses a proximal portion of the distal clamping portion **712** of the clip grasping structure **702** within the hypotube **700**, which also draws the prongs **706** and **708** closer together, thereby causing the distal clamping portion **712** of the clip grasping structure **702** to secure an interventional implant **220**. The elongated cutter **704** may then be advanced relative to the guide catheter **105**, hypotube **700** and clip grasping structure **702** to extend beyond the distal end **115** of the guide catheter **105** and engage with and cut the leaflet tissue. More specifically, the distal end of cutter **704** may be advanced through the first orifice **222** or the second orifice **224**, and then the system can be rotated about its central axis, causing cutter **704** to travel in an arcuate path around the interventional device **220**, slicing through the leaflet tissue located adjacent to the interventional device **220** as it travels from one orifice to the other. Of course, the cutter **704** could also be advanced through the leaflet tissue and then the system could be rotated back and forth until the entire cut from one orifice to the other orifice is complete. In addition, with this embodiment it is possible to cut through the tissue of both leaves of the mitral valve, thereby completely separating the interventional device **220** from the mitral valve. In that case, as shown in FIG. **7**C, the hypotube **700** and distal clamping portion **712** of the clip grasping structure **702** can be retracted in a proximal direction relative to the guide catheter and into the distal end **115** of the guide catheter **105**, thereby securing interventional implant **220**. FIG. **7**D shows the cut portion **610** of the anterior mitral leaflet **215**, however the present invention is not limited to the positioning or shape of cut.

Finally, reference is next made to FIGS. **8**A-**8**C, which illustrate still yet another embodiment of a cutting mechanism for use in detaching an interventional device from one or more leaflets of a mitral valve. In this embodiment, cutting mechanism **130***f* can include an inner catheter or hypotube **800** and a flexible stabilizing rod or dilator **802**. As shown, stabilizing rod **802** extends through the inner lumen of hypotube **800** and can be selectively advanced and/or retracted relative to hypotube **800**. The guide catheter is not shown in FIGS. **8**A and **8**C, but it should be understood that cutting mechanism **130***f* is intended for use within a suitable guide catheter to position the distal end of cutting mechanism **130***f* above the interventional device **220**. Once properly positioned, stabilizing rod **802** can be advanced relative to the guide catheter and hypotube **800** so as to extend beyond the distal end of hypotube **800** by a predetermined distance. With the stabilizing rod **802** advanced, the entire system can be advanced distally, until the distal end of stabilizing rod **802** engages the fold of leaflet tissue located opposite the interventional clip **220**, which fold is created when the posterior and anterior leaflets **210** and **215** are brought and held together by interventional device **220**.

As shown in FIGS. **8**A and **8**C, hypotube **800** can include one or more lumens formed in the wall of hypotube **800**, such as lumen **804** shown in FIG. **8**A and lumens **804** and **806** shown in FIG. **8**B. As illustrated, each lumen **804** and **806**, extends from the proximal end of hypotube **800** and then exits at an angle laterally outwardly from the side wall of hypotube **800** near the distal end of hypotube **800**. Positioned within each lumen **804** and **806** is an elongated cutter **808** and **810**. Each cutter **808** and **810** preferably terminates in a cutting blade portion **812** and **184** at its distal end, and can be selectively advanced and retracted relative to hypotube **800**. As shown in FIGS. **8**A and **8**C, when cutters **808** and **810** are advanced to extend beyond hypotube **800**, the distal ends of cutter **808** and **810** splay radially outwardly at an angle relative to the central axis of hypotube **800**.

In the case of the embodiment shown in FIG. **8**A, once the distal end of the stabilizing rod **802** is firmly engaged against the leaflet fold opposite the interventional device **220** as described above, then cutter **808** can be advanced to extend out and beyond the end of hypotube **800**. In some instances, the advancement of cutter **808** will also extend through one of the orifices **210** or **215**, but it may also be necessary to further advance hypotube **800** relative to stabilizing rod **802** until cutter **808** extends into orifice **222** or **224**. At that point, hypotube **800** alone or hypotube **800** and the guide catheter **105** together can be rotated around stabilizing rod **802**, thereby causing the cutting blade **812** of cutter **808** to slice through the leaflet tissue in an arcuate path around interventional device **220** as schematically illustrated in FIG. **8**B. With a single cutter **808**, a rotation of 180 degrees should be sufficient to detach one leaflet from the interventional device **220**, and a rotation of 360 degrees should be sufficient to detach both leaflets from the interventional device. As will be appreciated, the embodiment shown in FIG. **8**C operates in a substantially similar manner, except that each cutter **808** and **810** can simultaneously be inserted through both orifices, and cutting of both leaflets can be completed with a single rotation of 180 degrees.

In describing the various embodiments above, the description may at times have explicitly discussed one particular mitral valve leaflet, such as anterior leaflet **215**. It should be understood and appreciated, however, that the invention is not intended to be limited to either specific leaflet, but instead can be used to cut either anterior leaflet **215**, posterior leaflet **210**, or both.

It should also be understood that the order of manipulation of components of the various embodiments as described above are provided as representative examples only, and changes in the order of manipulation that may be readily understood by those skilled in the art are intended to be encompassed within the scope of this disclosure.

Further still, in addition to the embodiments described above, it should also be understood that individual components from one embodiment could also be combined with and/or substituted for a comparable component described in a different embodiment. For example, the stabilizing rod **802** disclosed in relation to the embodiment shown in FIGS. **8**A-**8**C could also be incorporated into one or more of the other embodiments, such as, for example, the embodiment illustrated in FIGS. **3**A-**3**D. Similarly, the cutting blades and structure shown in FIGS. **8**A and **8**C could be substituted in place of the cutting blade shown in FIGS. **7**A-**7**C, or vice versa.

Similarly, while many of the embodiments discussed above contemplate mechanical cutting of leaflet tissue by means of sharpened edges of a cutting element, it should be further understood that such embodiments could also be adapted to include suitable electrical connections between the cutting element and a source of electrosurgical energy so that such cutting elements may accomplish cutting of tissue by mechanical cutting, by the application of electrosurgical energy to surrounding tissue through the cutting element, or by a combination of both.

Also, with any or all of the foregoing embodiments, one or more components of the leaflet cutting system can also include one or more radiopaque and/or echogenic markers to aid in the visualization of such components during a procedure. For example, one or more radiopaque and/or echogenic markers can be provided on the distal end **115** and/or the steerable portion **117** of the guide catheter **105**. Similarly, one or more radiopaque and/or echogenic markers can also be provided on various components of the different embodiments of the cutting mechanisms described above, including, but not limited to such markings being provided on the distal ends of the inner catheter, hypotube, clip grasping structures, cutting blades, stabilizing rods, etc.

One skilled in the art will appreciate that the present invention is not limited to use within the mitral valve. The cardiac valve could also be the tricuspid aortic, pulmonic valve, etc. More generally, the embodiments described herein may be applied in other implementations involving removal of a previously implanted or deployed device from tissue. Further, although figures show the guide catheter **105** extending through the interatrial septum **200**, the present invention is not limited to use via a transseptal approach. Any suitable delivery approach may be used, including transfemoral, radial, transjugular, or transapical.

Following are some further example embodiments of the invention. These are presented only by way of example and are not intended to limit the scope of the invention in any way.

Embodiment 1. A system for cutting leaflet tissue at a cardiac valve, comprising a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve, a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve, and a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to cut a portion of leaflet tissue of the cardiac valve.

Embodiment 2. The system of embodiment 1, wherein the at least one control is further configured to provide selective actuation of the cutting mechanism.

Embodiment 3. The system in any of embodiments 1 to 2, wherein the guide catheter is introduced transseptally.

Embodiment 4. The system in any of embodiments 1 to 3, further comprising an indeflator attached to the guide catheter, the indeflator configured to hold the leaflet in place by applying negative pressure.

Embodiment 5. The system in any of embodiments 1 to 4, wherein the guide catheter is U-shaped.

Embodiment 6. The system in any of embodiments 1 to 5, wherein: the cutting mechanism comprises two blades joined at a pivot point, wherein the two blades each comprise a cutting edge disposed on an outside surface of each of the two blades and the two blades are oriented and shaped such that they cut the portion of leaflet tissue of the cardiac valve in a predefined arc; and the cutting mechanism is configured to extend from the distal end of the guide catheter.

Embodiment 7. The system in any of embodiments 1 to 6, wherein: the cutting mechanism comprises a hypotube with a sharped end that terminates in a point, wherein the sharpened end is configured to cut the portion of leaflet tissue of the cardiac valve in a predefined arc; and the cutting mechanism is configured to extend from the distal end of the guide catheter.

Embodiment 8. The system in any of embodiments 1 to 7, wherein: the cutting mechanism comprises a first and a second delivery catheter, each comprising a proximal end and a distal end; each of the first and second delivery catheters comprises a rotatable paddle joined to the delivery catheter by a joint at the distal end of the delivery catheter; each of the first and second delivery catheters comprises a gripping mechanism rotatable around the joint and positioned between the first delivery catheter and the rotatable paddle of the first delivery catheter, and the second delivery catheter and the rotatable paddle of the second delivery catheter; and a wire extending from the joint of the first delivery catheter to the joint of the second delivery catheter.

Embodiment 9. The system in any embodiments 1 to 8, wherein: the first delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the first delivery catheter into a first orifice of the cardiac valve, the gripping mechanism of the first delivery catheter and the rotatable paddle of the first delivery catheter are configured to secure leaflet tissue therebetween, the second delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the second delivery catheter into a second orifice of the cardiac valve, and the gripping mechanism of the second delivery catheter and the rotatable paddle of the second delivery catheter are configured to secure leaflet tissue therebetween.

Embodiment 10. The system in any of embodiments 1 to 9, wherein the gripping mechanism of each of the first and second guide catheters secures the leaflet tissue on an atrial side, and the rotatable paddle of each of the first and second guide catheters secures the leaflet tissue on a ventricular side.

Embodiment 11. The system in any of embodiments 1 to 10, wherein: the wire is configured to rotate around a hub within the joint of each of the first and second delivery catheters; and the wire configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated around the hub within the joint of each of the first and second delivery catheters.

Embodiment 12. The system in any of embodiments 1 to 11, further comprising a clip grasping structure comprising an elongated portion and a distal clamping portion; wherein: the distal clamping portion of the clip grasping structure is extendable distally past the distal end of the guide catheter and configured to secure an interventional implant that approximates adjacent leaflets of the cardiac valve; and the distal clamping portion of the clip grasping structure is retractable proximally into the distal end of the guide catheter with the secured interventional implant.

Embodiment 13. The system in any of embodiments 1 to 12, wherein the distal clamping portion of the clip grasping structure comprises two prongs.

Embodiment 14. The system in any of embodiments 1 to 13, wherein: the cutting mechanism comprises a cutting wire comprising an elongated portion that terminates in a distal loop portion, the cutting wire extends from the handle to the distal end of the guide catheter, and the distal clamping portion of the clip grasping structure extends through the distal loop portion of the cutting wire when extending distally past the distal end of the guide catheter and retracting proximally into the distal end of the guide catheter with the secured interventional implant.

Embodiment 15. The system in any of embodiments 1 to 14, wherein: the cutting wire is configured to selectively provide radio frequency current energy, and the distal loop portion of the cutting wire is configured to cut the portion of leaflet tissue when the distal clamping portion of the clip grasping structure retracts proximally into the distal end of the guide catheter with the secured interventional implant.

Embodiment 16. The system in any of embodiments 1 to 15, wherein the cutting wire is configured to detach the interventional implant from the surrounding leaflet tissue.

Embodiment 17. The system in any of embodiments 1 to 16, further comprising a hypotube routable through the guide catheter and able to be positioned at the distal end of the guide catheter, wherein the clip grasping structure is further routable through the hypotube and able to be positioned at a distal end of the hypotube.

Embodiment 18. The system in any of embodiments 1 to 17, wherein the hypotube is configured to extend from the distal end of the guide catheter when the distal clamping portion of the clip grasping structure is extended distally past the distal end of the guide catheter such that the hypotube encloses a proximal portion of the distal clamping portion of the clip grasping structure within the hypotube, thereby causing the distal clamping portion of the clip grasping structure to secure an interventional implant.

Embodiment 19. The system in any of embodiments 1 to 18, wherein the cutting mechanism comprises an elongated cutter configured to extend from the distal end of the guide catheter and rotate around a horizontal arc, thereby cutting the portion of leaflet tissue.

Embodiment 20. The system in any of embodiments 1 to 19, wherein the elongated cutter is configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated around the horizontal arc.

Embodiment 21. A method of cutting leaflet tissue at a cardiac valve within a body, comprising positioning a guide catheter, having a proximal and a distal end such that the distal end of the guide catheter is positioned at a cardiac valve, routing a cutting mechanism through the guide catheter such that the cutting mechanism extends to the distal end of the guide catheter, wherein the cardiac valve is associated with an interventional implant that approximates adjacent leaflets of the cardiac valve, and a cutting mechanism extends from the guide catheter; and actuating the cutting mechanism to cut at a portion of least one leaflet of the approximated adjacent leaflet.

The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

We claim:

1. A system for severing leaflet tissue at a cardiac valve, comprising:
- a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve;
- a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve;
- a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to sever a portion of leaflet tissue of the cardiac valve, wherein the cutting mechanism comprises:
  - a hypotube having a proximal end and a distal end with a lumen extending therebetween;
  - a tapering blade portion extending distally from a portion of the circumference of the distal end of the hypotube, the blade portion terminating at a distal most point longitudinally aligned with a first edge along the circumference of the hypotube;
    - wherein arcuate sharpened outer edges extend from the distal most point, the sharpened edges tapering proximally and radially outward along a circumference of the hypotube, the sharpened edges
    - terminating at a second edge diametrically opposed from the first edge; and
  - wherein the cutting mechanism is configured to sever the portion of leaflet tissue of the cardiac valve in a predefined arc around an interventional implant.

2. The system of claim **1**, wherein the at least one control is further configured to provide selective actuation of the cutting mechanism.

3. The system of claim **1**, wherein the guide catheter is introduced transseptally.

4. The system of claim **1**, further comprising an indeflator attached to the guide catheter, the indeflator configured to hold the leaflet in place by applying negative pressure.

5. The system of claim **1**, wherein the guide catheter is U-shaped.

6. The system of claim **1**, wherein:
- the cutting mechanism comprises two blades joined at a pivot point, wherein the two blades each comprise a cutting edge disposed on an outside surface of each of the two blades and the two blades are oriented and shaped such that they sever the portion of leaflet tissue of the cardiac valve in a predefined arc; and
- the cutting mechanism is configured to extend from the distal end of the guide catheter.

7. The system of claim **1**, wherein the cutting mechanism is configured to extend from the distal end of the guide.

8. The system of claim **1**, wherein:
- the cutting mechanism comprises a first and a second delivery catheter, each comprising a proximal end and a distal end;
- each of the first and second delivery catheters comprises a rotatable paddle joined to the delivery catheter by a joint at the distal end of the delivery catheter;
- each of the first and second delivery catheters comprises a gripping mechanism rotatable around the joint and positioned between the first delivery catheter and the rotatable paddle of the first delivery catheter, and the second delivery catheter and the rotatable paddle of the second delivery catheter; and
- a wire extending from the joint of the first delivery catheter to the joint of the second delivery catheter.

9. The system of claim **8**, wherein:
- the first delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the first delivery catheter into a first orifice of the cardiac valve;
- the gripping mechanism of the first delivery catheter and the rotatable paddle of the first delivery catheter are configured to secure leaflet tissue therebetween;
- the second delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the second delivery catheter into a second orifice of the cardiac valve; and
- the gripping mechanism of the second delivery catheter and the rotatable paddle of the second delivery catheter are configured to secure leaflet tissue therebetween.

10. The system of claim **9**, wherein the gripping mechanism of each of the first and second guide catheters secures the leaflet tissue on an atrial side, and the rotatable paddle of each of the first and second guide catheters secures the leaflet tissue on a ventricular side.

11. The system of claim **9**, wherein:
- the wire is configured to rotate around a hub within the joint of each of the first and second delivery catheters; and
- the wire configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated around the hub within the joint of each of the first and second delivery catheters.

12. The system of claim **1**, further comprising a clip grasping structure comprising an elongated portion and a distal clamping portion; wherein:
- the distal clamping portion of the clip grasping structure is extendable distally past the distal end of the guide catheter and configured to secure an interventional implant that approximates adjacent leaflets of the cardiac valve; and
- the distal clamping portion of the clip grasping structure is retractable proximally into the distal end of the guide catheter with the secured interventional implant.

13. The system of claim **12**, wherein the distal clamping portion of the clip grasping structure comprises two prongs.

14. The system of claim **12**, wherein:
- the cutting mechanism comprises a cutting wire comprising an elongated portion that terminates in a distal loop portion;
- the cutting wire extends from the handle to the distal end of the guide catheter; and
- the distal clamping portion of the clip grasping structure extends through the distal loop portion of the cutting wire when extending distally past the distal end of the guide catheter and retracting proximally into the distal end of the guide catheter with the secured interventional implant.

15. The system of claim **14**, wherein:
- the cutting wire is configured to selectively provide radio frequency current energy; and
- the distal loop portion of the cutting wire is configured to sever the portion of leaflet tissue when the distal

clamping portion of the clip grasping structure retracts proximally into the distal end of the guide catheter with the secured interventional implant.

16. The system of claim **14**, wherein the cutting wire is configured to detach the interventional implant from surrounding leaflet tissue.

17. The system of claim **12**, further comprising a hypotube routable through the guide catheter and able to be positioned at the distal end of the guide catheter, wherein the clip grasping structure is further routable through the hypotube and able to be positioned at a distal end of the hypotube.

18. The system of claim **17**, wherein the hypotube is configured to extend from the distal end of the guide catheter when the distal clamping portion of the clip grasping structure is extended distally past the distal end of the guide catheter such that the hypotube encloses a proximal portion of the distal clamping portion of the clip grasping structure within the hypotube, thereby causing the distal clamping portion of the clip grasping structure to secure an interventional implant.

19. The system of claim **17**, wherein the cutting mechanism comprises an elongated cutter configured to extend from the distal end of the guide catheter and rotate around a horizontal arc, thereby cutting the portion of leaflet tissue.

20. The system of claim **19**, wherein the elongated cutter is configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated the horizontal arc.

21. A method of cutting leaflet tissue at a cardiac valve within a body, comprising:

positioning a guide catheter, having a proximal and a distal end such that the distal end of the guide catheter is positioned at a cardiac valve;

routing a cutting mechanism through the guide catheter such that the cutting mechanism extends to the distal end of the guide catheter, wherein the cardiac valve is associated with an interventional implant that approximates adjacent leaflets of the cardiac valve, and a cutting mechanism extends from the guide catheter, and the cutting mechanism comprises:

a hypotube having a proximal end and a distal end with a lumen extending therebetween;

a tapering blade portion extending distally from a portion of the circumference of the distal end of the hypotube, the blade portion terminating at a distal most point longitudinally aligned with a first edge along the circumference of the distal end of the hypotube;

wherein arcuate sharpened outer edges extend from the distal most point, the sharpened outer edges tapering proximally and radially outward along a circumference of the hypotube, terminating at a second edge diametrically opposed from the first edge, wherein the cutting mechanism is configured to sever the portion of leaflet tissue of the cardiac valve in a predefined arc; and

actuating the cutting mechanism to sever at a portion of at least one leaflet of the approximated adjacent leaflet.

* * * * *